



YOUR KINDLE NOTES FOR:

Seveneves: A Novel

by Neal Stephenson

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262 Highlights | 7 Notes

Highlight (Yellow) | Location 704

a torus—a donut-shaped habitat about forty meters in diameter—was constructed and made to spin like a merry-go-round, creating a small amount of simulated gravity.

Highlight (Yellow) | Location 777

overwhelming majority of people called him Doc Dubois now, because he was on TV a lot, and that was how the talk show hosts and the network anchormen introduced him. His job on TV was to explain science to the general public and, as such, to act as a lightning rod for people who could not accept all the things that science implied about their worldview and their way of life, and who showed a kind of harebrained ingenuity in finding ways to refute it.

Highlight (Yellow) | Location 789

being Doc Dubois, the affable science popularizer of television and of four million Twitter followers. Doc Dubois knew how to size up his audience. He knew that self-made tech zillionaires liked to argue, that Pasadena aristocracy didn't, and that society wives liked to be lectured to, as long as the lectures were brief and funny. And he knew that his job was to charm these people, nothing more, so that they could later be handed off to professional fund-raisers. He was going back to the bar for another glass of the pinot noir,

Highlight (Yellow) | Location 827

When the meeting began at A+0.0.4 (zero years, zero days, and four hours since the Agent had acted upon the moon),

Highlight (Yellow) | Location 831

“Primordial singularity”

Highlight (Yellow) | Location 837

“A small black hole.” “Why ‘primordial’?” “Most black holes are formed when stars collapse,” Ivy said. “But there’s a theory that some of them were created shortly after the Big Bang. The universe was lumpy. Some of

the lumps might have been dense enough to undergo gravitational collapse. They could form black holes that instead of weighing what a star weighs could be a lot smaller.” “How small?” “I don’t think there’s a lower limit. But the point is that one of them could zip through space invisibly and punch all the way through a planet and out the other side. There

Highlight (Yellow) | Location 849

Like a bullet through an apple.”

Highlight (Yellow) | Location 894

Uppity Little Shitkicker was a name that had been hung on Dinah when she had first arrived at private school and, during a soccer scrimmage, intercepted a pass meant for a girl from New York. Straight Arrow Bitch had been bestowed on Ivy at Annapolis when she had declined to take part in a drinking game during a tailgate party. The ULS/SAB dynamic was a thing that Dinah and Ivy exploited in meetings,

Highlight (Yellow) | Location 949

Because making observations calmed people down. The moon had broken up into seven large pieces, which inevitably became known as the Seven Sisters, and an uncountable number of smaller ones. Gradually the big ones acquired names. Doc Dubois was responsible for many of these. He gave them descriptive names that wouldn’t scare people. It wouldn’t do to call them Nemesis or Thor or Grond. So instead it was Potatohead, Mr. Spinny, Acorn, Peach Pit, Scoop, Big Boy, and Kidney Bean. Doc Dubois would point those out and then draw attention to the way they moved. This was governed entirely by Newtonian mechanics. Each piece of the moon attracted every other piece more or less strongly depending on its mass and its distance. It could be simulated on a computer quite easily. The whole rubble cloud was gravitationally bound.

Highlight (Yellow) | Location 956

loose huddle of rocks. Sometimes they banged into one another. Eventually they would stick together and the moon would begin to re-form. Or at least that was the theory until the star party that they threw in the middle of the Caltech campus at A+0.7.0, exactly one week after the event.

Highlight (Yellow) | Location 1003

“WE NEED TO STOP ASKING OURSELVES WHAT HAPPENED AND START talking about what is going to happen,” Dr. Harris said to the president of the United States, her science advisor, the chairman of the Joint Chiefs of Staff, and about half of the Cabinet.

Highlight (Yellow) | Location 1014

The slide was a montage of a snowball rolling down a hill, a fuzzy bacterial culture growing in a petri dish, a mushroom cloud, and other seemingly unrelated phenomena. “What do these all have in common? They are exponential,” he said. “The word gets tossed around a lot by people who use it to mean anything that’s getting

big fast. But it has a specific mathematical meaning. It means any process where the more it happens, the more it happens. The population explosion. A nuclear chain reaction. A snowball rolling down a hill, whose speed of growth is pegged to how much it's grown." He clicked through another slide showing plots of exponential curves on a graph, then to an image of the moon's eight pieces. "When the moon had only one piece, the probability of a collision was zero," he said.

Highlight (Yellow) | Location 1033

"It could be that a hundred years will pass before we go from eight chunks to nine chunks," Doob said, nodding at him, "but four days ago I got worried that it might be one of those things that looks more like an explosion. So my grad students and I have been crunching some numbers. Building a mathematical model of the process that we can use to get a handle on the time scale." "And what are your results, Dr. Harris? I assume you have some, or else you wouldn't be here." "The good news is that the Earth is one day going to have a beautiful system of rings, just like Saturn. The bad news is that it's going to be messy."

Highlight (Yellow) | Location 1057

"A day or two after the White Sky event will begin a thing I am calling the Hard Rain. Because not all of those rocks are going to stay up there. Some of them are going to fall into the Earth's atmosphere." He turned the projector off. This was an unusual move, but it snapped them all out of PowerPoint hypnosis and forced them to look at him. The aides in the back of the room were still thumbing their phones, but they didn't matter. "By 'some,'" Doob said, "I mean trillions." The room remained silent. "It is going to be a meteorite bombardment such as the Earth has not seen since the primordial age, when the solar system was formed," Doob said. "Those fiery trails we've been seeing in the sky lately, as the meteorites come in and burn up? There will be so many of those that they will merge into a dome of fire that will set aflame anything that can see it. The entire surface of the Earth is going to be sterilized. Glaciers will boil. The only way to survive is to get away from the atmosphere. Go underground, or go into space."

Highlight (Yellow) | Location 1068

"We will have to do both," the president said. "Go into space, and underground. Obviously the latter is easier." "Yes." "We can get to work building underground bunkers for . . ." and she caught herself before saying something impolitic. "For people to take refuge in."

Highlight (Yellow) | Location 1072

The chairman of the Joint Chiefs of Staff said, "Dr. Harris, I'm an old logistics guy. I deal in stuff. How much stuff do we need to get underground? How many sacks of potatoes and rolls of toilet paper per occupant? I guess what I'm asking is, just how long is the Hard Rain going to last?" Doob said, "My best estimate is that it will last somewhere between five thousand and ten thousand years."

Highlight (Yellow) | Location 1076

"NONE OF YOU WILL EVER STAND ON TERRA FIRMA, TOUCH YOUR loved ones, or breathe the atmosphere of your mother planet again," the president said. "That is a terrible fate. And yet it is a better fate

than seven billion people trapped on the Earth's surface can hope for. The last ship home has sailed. From now on, launch vehicles will rise up into orbit, but they will not go back for ten thousand years." The twelve men and women in the Banana sat in silence. Like the destruction of the moon itself, it was too big a thing for them to take in, too large for human emotion to get around.

Highlight (Yellow) | Location 1090

"We have signals intelligence suggesting that the Chinese figured it out a day before we did," the president said, "and the British, the Indians, the French, Germans, Russians, Japanese—all of their scientists are coming to more or less the same conclusions." "Two years?" Dinah piped up. Her voice was hoarse, broken. Everyone looked at her. "Until the White Sky?" "People seem to be converging on that figure, yeah," Dr. Harris said. "Twenty-five months, plus or minus two."

Highlight (Yellow) | Location 1099

"We are beginning to talk to our counterparts in other spacefaring nations about creating an ark," the president said. "A repository of the entire genetic heritage of the Earth. We have two years to build it. Two years to get as many people and as much equipment as we can into orbit. The nucleus of that ark is going to be Izzy."

Highlight (Yellow) | Location 1103

Things had to be humanized, to be given cute names. All the terrified kids down there who knew they were going to die would have to watch upbeat videos about how Izzy was going to carry the legacy of the dead planet through the Hard Rain. They would take out their crayons and draw cartoon pictures of Izzy with a torus halo and a big rock on her ass and a little anthropomorphic smiley face on the side of the Zvezda Service Module.

Highlight (Yellow) | Location 1228

she saw the head and arm of a single cosmonaut, whom she vaguely recognized as Maxim Koshelev. He was embedded in a nearly solid mass of vitamins. "Vitamins" was a term of art used by spaceflight geeks to mean any small, lightweight stuff of extraordinary value. Microchips, medicine, spare parts, ukuleles, biological samples, soap, and food all fell under the general heading of "vitamins."

Highlight (Yellow) | Location 1249

So, both Bolor-Erdene and Maxim had ridden in the orbital module, which was unprecedented; humans were supposed to ride only in the reentry module aft of it. It would have been indiscreet to point this out, but those two, by riding up front, had signed up for a one-way journey that could have turned into a suicide mission had anything gone wrong. The orbital module was jettisoned during the reentry process, and burned up in the atmosphere. Only the passengers in the reentry module could even theoretically make it back alive.

Highlight (Yellow) | Location 1345

Buckyball seems to be using a similar trick." "You nailed the name. We do in fact call those Buckies. Technically speaking, it's a thing called a—" "Tensegrity."

Highlight (Yellow) | Location 1355

“I’m not a normal engineer. I’ve been doing biomimetics—which is what you are doing—for

Highlight (Yellow) | Location 1382

If the task had fallen to Dubois Jerome Xavier Harris, Ph.D., he would have said something like this: Look, everybody dies. Of the seven billion people now living on Earth, basically all will be dead a hundred years from now—most a lot sooner. No one wants to die, but most calmly accept that it’s going to happen. A person who died two years from now in the Hard Rain would be no deader than someone who died seventeen years from now in a car crash. The only thing that had changed now was that everyone knew the approximate time and manner of their death. And knowing that, they could make preparations. Some of those were internal: making your peace with your God.

Highlight (Yellow) | Location 1475

Special machines would be sent up on the Cloud Ark, machines that could take those digital records and turn them back into functioning DNA and embed them in living cells, so that giraffes and sequoias and whales could be reconstituted from raw elements at some point down the road, perhaps thousands of years in the future.

Highlight (Yellow) | Location 1504

“We ask every village, town, city, and district to perform a Casting of Lots and to choose two young persons, a boy and a girl, as candidates for training and inclusion in the crew of the Cloud Ark. We do not seek to impose any rules or procedures upon how the selection is made. Our objective is to preserve, as best we can, the genetic and the cultural diversity of the human race.

Highlight (Yellow) | Location 1508

The statement was subtly self-contradictory. Clarence was saying that they were not going to impose any rules. But they had already done so by insisting that there had to be both a boy and a girl. They knew perfectly well that many cultures would have trouble with that.

Highlight (Yellow) | Location 1567

Research on the long-term effects of spaceflight suggested that women were less susceptible to radiation damage than men. They were smaller on average, requiring less space, less food, less air. And sociological studies pointed to the idea that they did better when crammed together in tight spaces for long periods of time.

Highlight (Yellow) | Location 1569

This was controversial, as it got into fraught topics of nature vs. nurture and whether gender identity was a social construct or a genetic program. But if you bought into the idea that boys had been programmed by Darwinian selection to run around in the open chucking spears at wild animals—something that every parent who had ever raised a boy had to take seriously—then it was difficult to envision a lot of them spending their lives in tin cans.

Highlight (Yellow) | Location 1695

Sublimation was essentially the same thing as evaporation, skipping the liquid phase; it just meant a process by which a solid, exposed to vacuum, gradually turned into vapor and disappeared. Ice tended to do this pretty quickly unless it was kept extremely cold. "So Izzy is going to lose water," Dinah said, "which is a scarce and valuable resource."

Highlight (Yellow) | Location 1768

Jamming more people and systems into the central body of the space station would just require more solar panels, more batteries, more radiators, and more plumbing and wiring to connect them all.

Highlight (Yellow) | Location 1769

the human factors: how to supply people with food, water, and clean breathable air, and how to recycle carbon dioxide and sewage.

Highlight (Yellow) | Location 1772

arklet, as the component ships were being called, would be small enough that it could be heaved into orbit on the top of a single heavy-lift rocket. It would draw power from a small, simple nuclear reactor fueled by isotopes so radioactive that they would throw off heat, and thereby generate electricity, for a few decades.

Highlight (Yellow) | Location 1835

air inside the innermost bag would get fouled with carbon dioxide as the occupant breathed, but this could be handled as it usually was on spaceships and submarines, by passing the air over a chemical such as lithium hydroxide that would absorb the CO₂. As long as a bit of oxygen was bled in to replace what was being used, the occupant would be fine.

Highlight (Yellow) | Location 1882

together in two weeks from existing technology.

Highlight (Yellow) | Location 1906

The normal up/down cycle of launch and reentry had, of course, been suspended. Things were supposed to go up, but not come down.

Highlight (Yellow) | Location 2051

It had all become Kremlinology. Back in the heyday of the Soviet Union, the only way for Westerners to guess what was going on there was to look at the lineup of dignitaries on Lenin's Tomb in the May Day parade, and riddle it out from the seating chart and who shook hands with whom. Now Dinah was doing the same thing with these three faces

Highlight (Yellow) | Location 2554

Sambo, a Soviet combat martial art with many similarities to jujitsu.

Highlight (Yellow) | Location 2573

“The closest analogy in the law for what I am here is the captain of a ship,” Ivy said. “Do you know the etiquette, Sean, for coming aboard a ship?” Sean calculated. “Commander Xiao,” he said, “I humbly and respectfully request permission to come aboard your ship.” “Permission granted,” she said. “And welcome aboard.”

Highlight (Yellow) | Location 2594

She hadn’t been following the stock market since the Crater Lake announcement, but she’d heard that it had not so much crashed as basically ceased to exist. The whole concept of owning stock didn’t really mean much anymore, at least if you thought of it as a store of value.

Highlight (Yellow) | Location 2635

The seven billion people angle. Seven billion who need to be kept happy, and docile, until the end. How do you do that? What’s the best way to calm down a scared kid, get them to go back to sleep? Tell them a story. Some shit about Jesus or whatever.”

Highlight (Yellow) | Location 2681

An acquaintance of his who had made a lot of money in hedge funds had dropped twenty-five million dollars on a twelve-day trip to the International Space Station aboard a Soyuz capsule.

Highlight (Yellow) | Location 2690

A week before the scheduled launch, he had flown to Baikonur, bringing his wife and kids with him, video crew in tow. They had watched in a certain amount of amazement as the launch vehicle, a fantailed Soyuz-FG, had been towed horizontally across the steppe on a special train, complete with smoke-belching locomotive, to the launch pad. And this really was little more than a pad, a concrete slab on the almost lunar surface of the Kazakh steppe with a few pieces of apparatus around it to hoist the rocket up off the train and pump fluids into it. The contrast with the NASA way of doing things was stark to the point of being somewhat hilarious.

Highlight (Yellow) | Location 2721

In a certain way it made no sense anymore to speak of anyone as being a student at a particular stage in a degree program. And yet people went on thinking this way,

Highlight (Yellow) | Location 2722

someone who has just been diagnosed with a terminal illness will go on getting up and going to work every morning, not so much out of habit as because the knowledge of impending doom makes them wish to assert an identity.

Highlight (Yellow) | Location 2724

He was tempted to park the SUV illegally, since, according to his calculations, the authorities were not likely to catch up with him and demand payment of the parking ticket before the end of the world, but it seemed that most of the people of Seattle were still obeying the rules and so he did likewise.

Highlight (Yellow) | Location 2730

This morning, other than getting up from his La-Z-Boy to give his dad a hug and tell him “I love you,” he didn’t have much more to say. Every parent of a teenager gets used to it: the moment in a child’s life when he or she decides that certain facts are just too much trouble to explain to Mom or Dad. The parents can’t, and needn’t, know every last little thing. They just have to accept this, be content with what they can glean on their own, and move on. Henry, of course, had passed through that veil some years ago. Doob had swallowed his pride and accepted it as every parent must.

Highlight (Yellow) | Location 2748

a direct hit and become part of a huge glass-lined crater.

Highlight (Yellow) | Location 2748

The truck didn’t stop soon enough and wrecked a wooden banister beside the front steps. Everyone laughed. The laughter had a curious tone, a mixture of childish delight with something darker, expectant of much worse to come. These kids were really adapting better than he was.

Highlight (Yellow) | Location 2756

Eventually someone went and got another box truck. Apparently the rental agency was letting them take them for free. Some day laborers wandered down the street from a home improvement center and helped pack. The home improvement market had gone bust.

Highlight (Yellow) | Location 2765

Henry was a sweet kid and Doob knew that when he woke up he would apologize. But Henry wasn’t a parent, and he didn’t understand that when you were, almost nothing was more satisfying than seeing your kid sleep.

Highlight (Yellow) | Location 2836

Not the meteorite itself, but ejecta: shrapnel hurled out from the impact site.

Highlight (Yellow) | Location 2867

“We talk about sending stuff to orbit as if orbit is a place, like Philadelphia, but it’s actually a lot of places, a lot of different ways to be in space. Any two objects in the universe can theoretically be in orbit around each other. “Most of the orbits that matter to us involve something tiny orbiting around something huge, like a satellite around the Earth, or the Earth around the sun. So, a quick way to label and classify orbits is according to ‘What is the huge thing in the middle?’ “If the huge thing in the middle is the Earth, we call it a geocentric orbit. If it’s the sun, it’s a heliocentric orbit. And so on. Since the moon broke up, we’ve mostly focused on geocentric orbits. The moon, back when it existed, used to be in such an orbit; it revolved around the Earth. Most of its pieces still remain in geocentric orbits. A small number of those just happen to intersect the Earth’s atmosphere. When that happens, we get a meteorite. “So much for Orbits 101.

Highlight (Yellow) | Location 2876

But keep in mind there can be different levels. So, the old Earth-moon system was, as a whole, revolving around the sun in a heliocentric orbit. And if you zoom way out and look at the entire Milky Way galaxy, you can see that our whole solar system is very slowly revolving around the black hole at its center, in a galactocentric orbit.”

Highlight (Yellow) | Location 2910

Our Heritage phrasing, a suddenly popular buzzword and hashtag meaning “whatever gets accomplished in the distant future by the descendants of the people who make it onto the Cloud Ark,” or, to put it bluntly, “the only reason to go on living for the next twenty-two months.”

Highlight (Yellow) | Location 2955

Izzy’s orbit is angled with respect to the equator. It has to be that way so Baikonur can launch to it—Baikonur is as far north as Seattle. But when you are doing interplanetary stuff, which is what Sean has in mind—basically, whenever you want to get out of low Earth orbit—you want to be in an orbit that’s closer to the equator. Because that’s pretty much where the rest of the solar system is—including the big chunk of ice that Sean wants to grab and bring back here.”

Highlight (Yellow) | Location 2982

“These are more like abstract mathematical points,” Dinah said. “They’re called the Lagrange points, or the libration points, and there’s five of them around every two-body system. Always in the same basic geometry. Two of them, L4 and L5, are way off to the sides. I’m not going to try to show you those

Highlight (Yellow) | Location 2989

Finally, she put a pill in between the “sun” and the “Earth” but much closer to the latter. “And this is—” “L1, by process of elimination,”

Highlight (Yellow) | Location 3003

L1 is four times farther away than the moon.” “Or what used to be the moon,” Luisa said under her breath. “Yeah, which means that in a few days Sean and his crew are going to be farther away from Earth than any humans who have ever lived.

Highlight (Yellow) | Location 3030

On Day 73, when Ivy had opened another meeting in the Banana with the words “ten percent,”

Highlight (Yellow) | Location 3033

The population had certainly gone into the triple digits at least a week before.

Highlight (Yellow) | Location 3079

when Sean and Larz had departed in the Drop Top on A+0.82. The plane-change maneuver took them to a rendezvous above the equator with Ymir. After more rendezvous operations extending over a week, and incorporating yet more payloads launched from Cape Canaveral as well as from private spaceports in New Mexico and West Texas, Ymir made a long burn of her main engine that placed her into a transfer orbit bound for L1. A few days after that, she beat the Apollo record for distance traveled from Earth.

Highlight (Yellow) | Location 3151

calculations at other universities around the world, were that the White Sky would happen on or about A+1.354, or one year and 354 days after the breakup of the moon; they were one-fifth of the way there).

Highlight (Yellow) | Location 3163

The only way to get one’s bearings, looking at a rendering of the complex, was by picking out the rugged and asymmetrical shape of Amalthea at one end and the two tori at the other. Those were “forward” and “aft,” respectively, and the axis between them was the basis for the traditional nautical directions of “port” and “starboard” as well as “zenith” and “nadir,” which were space lingo for basically “up away from Earth” and “down toward Earth.”

Highlight (Yellow) | Location 3171

Felt-tip markers, always a scarce resource even on Earth, became objects of great value as people used them to mark directions on the walls of hamster tubes and habitat modules.

Highlight (Yellow) | Location 3233

Izzy had been like a kite: all surface area, no mass. In technical terms, it had had a low ballistic coefficient: a way of saying that it was strongly affected by what little atmosphere there was. Once Amalthea had been

attached, it was like a kite with a big rock strapped to it. It had a high ballistic coefficient. The rock's momentum bulled through the evanescent atmosphere and led to much slower orbital decay. But by the same token, when it came time to reboost Izzy's orbit, a longer burn and a larger amount of propellant were needed in order to accelerate all of that iron and nickel.

Highlight (Yellow) | Location 3250

the heat rejection complex waxed enormous and ended up growing generally aftward, like an empennage—the feathers on the butt of an arrow.

Highlight (Yellow) | Location 3259

The radiators were, in essence, a gigantic exploit in zero-gravity plumbing. The excess heat had to be collected from where it was produced (mostly, the inhabited and pressurized parts of Izzy) and transported to where it could be gotten rid of (the “empennage” growing to aft). The only plausible way of doing this was by using a fluid, pumping it around a loop, heating it up at one end and cooling it off at the other.

Highlight (Yellow) | Location 3298

Connecting segments of beam into more complex structures, such as trusses and scaffolding, was a little more difficult. Welding in space, while possible, was complicated, and there wasn't enough equipment. Instead they ended up using Tinkertoy-like connectors, again mass-produced by the Chinese, into which the ends of the triangular beams could be inserted, then tightened down using screws. At first many of these were shipped up in bulk from the ground, but on A+0.247 they took delivery of a 3-D printer that had been optimized to make more of them, with options for modifying the angle at which the beams would be inserted. This gave them the ability to design and build trusses on the fly, which was not possible with the mass-produced connectors.

Highlight (Yellow) | Location 3303

And as a last resort, Fyodor had an electron-beam welding machine that would work in zero gravity and a vacuum, undoubtedly the most expensive welder ever made, a marvel of Russian ingenuity, and he had trained Vyacheslav to use it. Vyacheslav then trained Tekla and two of the other spacewalkers, who set up a job queue and took turns drifting around Izzy's increasingly complex structure tacking down a weld here and a weld there.

Highlight (Yellow) | Location 3354

You have multiple core competencies with surprisingly minimal Venn.

Highlight (Yellow) | Location 3367

there were two questions in his head at the same time: whether he would lie, and whether he could lie. As to the first question, was it okay for him to lie if it would make billions of people go to their deaths a little happier?

Highlight (Yellow) | Location 3373

One of the ice cubes in Doob's glass let out a little pop as it underwent thermal fracturing.

Highlight (Yellow) | Location 3382

water was always a good thing to have in space: you could drink it, grow crops with it, use it for radiation shielding, split it into hydrogen and oxygen for rocket fuel, or pipe it through hoses to radiate excess heat into space. All of which was quite true, but sort of begged the question. It was so blindingly obvious that NASA must have thought of it already. What additional demand for water was Sean Probst seeing that NASA had failed to notice, or turned a blind eye to? Later Doob had figured it out based on background conversations with people at Arjuna and scuttlebutt reaching him through friends working on the planning of the Cloud Ark. It was all about propellant. The Cloud Ark would have to burn a lot of it.

Highlight (Yellow) | Location 3433

They could drive down to Fairbanks any time they feel like it, and from there go anywhere in the world. Check out the pyramids. See the Mona Lisa. Visit old friends and family. Instead they're up in the most godforsaken place I've ever seen, doing what?" "Prepping?" Ivy said. "That's the only thing I can think," Dinah said. "Prepping for a five- to ten-thousand-year stay." "They're not the only ones," Ivy said.

Highlight (Yellow) | Location 3439

"Mixed in with the stuff you'd expect from a fiancé—which is none of your business—he asks me questions about things like the comparative merits of lithium versus sodium hydroxide scrubbers. He requests copies of Luisa's PDFs about the sociology of persons confined in small places for long periods." "He can't think you're not going to notice that." "Sure. I'm going to read between the lines." "What do you suppose he's thinking?" "Well," Ivy said, "he does have sole authority over a huge submarine designed to ride out global thermonuclear warfare. And when the United States ceases to exist, I guess there'll be no one above him, chain-of-command-wise. What's a commander to do?" "But how would it work?" "I think a lot depends," Ivy said, "on whether the oceans boil dry. If I were him, I'd make for the Marianas Trench and keep my fingers crossed."

Highlight (Yellow) | Location 3450

"Staying alive in space is going to be a piece of cake, remember?" "Oh yeah, sorry. I forgot . . ." To put on my makeup. "It would present some fascinating challenges," she corrected herself, switching to her best NASA PR voice. "I think it's like what we are doing," Ivy said. "You have to break it down into a lot of little things and solve them one at a time, or you get overwhelmed." "Is that what we're doing?" "Yeah." Ivy rolled her eyes.

Highlight (Yellow) | Location 3465

During the first few months after Zero, no one had paid much attention to long-term health issues. Everyone was going to die. Scouts were showing up dead on arrival. It had been all emergency, all the time. But during the months of hamster tube building and structural consolidation, the life scientists had been quietly having their

say. This wasn't the first time Dinah had been nudged in recent weeks about her failure to spend more time in the simulated gravity field of T2. "It's just hard to go back and forth between gravity and no gravity," Dinah said. "It makes me barf. And none of my stuff is in T2."

Highlight (Yellow) | Location 3473

"Whatever you're doing here," Ivy continued, "you could do from a cabin in T2, where the gravity would build your bones."

Highlight (Yellow) | Location 3494

What's your role on the reality TV show, Ivy?" Ivy stared at her coolly. "I'm the uptight bitch who can't handle it." "Oh." "To American viewers, I'm not fully American. To Chinese viewers, I'm a banana." "I'm sorry, Ivy." "That's the bad news." "Okay, and what is the good news?" "All the people saying mean things about me on the Internet are gonna be dead in four hundred and thirty-three days," she said, deadpan. Okay. It was an example of that dark humor thing.

Highlight (Yellow) | Location 3525

While he'd been flying back to L.A., a hunk of it the size of an oil tanker had slammed into the Indian Ocean and kicked up a tsunami that had killed forty thousand people on the west coast of India.

Highlight (Yellow) | Location 3538

He knew that they would shatter into an uncountable number of fragments and become the White Sky and then the Hard Rain. But what was the final distribution of sizes going to be, how many big ones and how many small? They had some models based on the simplifying assumption that all moon rock was basically the same, but clearly that wasn't true.

Highlight (Yellow) | Location 3542

Peach Pit was much denser than the other bits, suggesting that it consisted mostly of iron as opposed to rock. The moon had had an iron core, but, relative to overall size, this was much smaller than the Earth's;

Highlight (Yellow) | Location 3560

Survival was the first imperative. Long-term strategy came next.

Highlight (Yellow) | Location 3562

The amount of iron in PP1 was for all practical purposes infinite. It would take humans many thousands of years to find uses for that much metal. But it was way up high. Hard to reach. And yet they had to reach it. And it was closer, easier to reach, than the Arjuna asteroids that Sean Probst was so excited about. Feeling an idea take shape in his head, like an iron core congealing deep in a moon, he put it on hold and forced himself to turn his attention to more immediate questions.

Highlight (Yellow) | Location 3570

He hadn't wanted to go, but once he had assented to the idea, he had begun wanting it more than anything, and he now feared that they would take it away from him. And if they sensed that fear, they could use it to control him. Better to overperform, to exceed expectations, to act like it was nothing at all.

Highlight (Yellow) | Location 3641

"So if anyone hands you anything, even if you have no idea what on God's green Earth it might be, look impressed, bow, take it carefully, admire it, and then hand it off to the helper kid."

Highlight (Yellow) | Location 3643

"People have been deputized to follow you around and help you carry all of the priceless national treasures that are going to be bestowed on you. They'll look after the stuff and bring it all back here to the chopper so you can keep your hands free for making those little bows and shaking hands with the king or whatever. As soon as we get back to the aircraft carrier, we'll throw it overboard."

Highlight (Yellow) | Location 3676

have come to the same conclusion," Doob said. "A pool of promising candidates is identified and then the choice is made from among them by some process which might be random—just so no one person carries the entire responsibility." "When you are a king you sometimes have such responsibilities whether you want them or not. In this case, though, I was able to involve some of the lamas. There are precedents for such a selection procedure in the way that certain reincarnate lamas are identified—the

Highlight (Yellow) | Location 3686

"When seven billion die, and only some thousands remain, where do the seven billion souls go?"

Highlight (Yellow) | Location 3692

"We don't believe in anything as simple as metempsychosis—the movement of an individual soul from one body to another."

Highlight (Yellow) | Location 3744

He now understood his job a lot more clearly. Why were the doomed people of Earth not going completely berserk? Oh, there had been some outbreaks of civil disorder, but for the most part people were taking it surprisingly calmly. It was because events like this were happening in every city and province with more than a few hundred thousand people, and they were being stage-managed well enough to convince people that the system was working. When he was a kid he had read the Greek myth of Theseus and the minotaur, which hinged on the premise that the people of Athens had somehow been persuaded to select seven maidens and seven boys by lot, every few years, and send them to Crete to serve as monster chow. This had always struck him as the

weakest point of what was otherwise a great yarn. Who would do that? Who would choose their kids by lot and send them to such a fate? The people of Bhutan, that was who. And the people of Seattle and of

Highlight (Yellow) | Location 3772

The first three arklets were launched “naked” and smooth, but the hundreds to follow would come up clothed in translucent fabric jackets, pleated and wrinkled during the passage through the atmosphere, protected beneath fiberglass fairings. Once in space the blanket would be inflated to form a flexible outer hull somewhat larger than the inner one. It was in that inter-hull space where food would be grown, making use of sunlight that would diffuse through the fabric. It was not clear yet whether each arklet could be self-sufficient in terms of food production—probably not—but growing some food was better than growing none. Having some green stuff on board helped reduce the load on the CO2 scrubbers, and having water between humans and space helped stop some incoming radiation.

Highlight (Yellow) | Location 3935

“Congratulations, Bolo One,” said the engineer down in Houston. “The first autonomously driven coupling of two spacecraft to create a rotating system for production of Earth-normal simulated gravity.”

Highlight (Yellow) | Location 3941

The sound of the cable reels changed as automatic brakes engaged to slow their unwinding and avoid a damaging jerk at the end. Then there was silence for a few moments, and then another thruster burn—longer, and directed laterally, to speed up the bolo’s rotational velocity. “Holy shit” was the only thing that Dinah could say for the first minute or two. They were experiencing one gee—Earth-normal gravity—for the first time in over a year. Markus, who’d only been in orbit for a few days, sounded great.

Highlight (Yellow) | Location 3992

“Look, you know my views on the Singularity. On uploading.” “I did blurb your book on the topic.” “Yes, thank you for that.” Tav was referring to the idea that the human brain could, in principle, be digitized and uploaded into a computer. That this would one day happen on a large scale. That it might actually have happened already—that we might all, in fact, be living in a giant digital simulation. Something occurred to Doob. “Is that why you were grilling the king about his views on reincarnation?” “That’s part of it,”

Highlight (Yellow) | Location 3997

“Look, all I’m saying is that if you’ve gone where I’ve already gone, in terms of thinking about that—” “If you’ve drunk the Singularity Kool-Aid, in other words?” Doob said. “Yeah, Doob, as you know I’ve already done, then you’ve already made a fundamental break with trying to be Nature Boy. I am never going to be Nature Boy. I believe that the human mind is almost infinitely malleable and that people are going to adjust, within days or weeks, to life on the Cloud Ark. We will simply turn into a different civilization altogether from the one we grew up in. Our whole idea of nature will be forgotten. And a thousand years from now, people will go on ‘camping trips’ that will consist of sleeping in arklets, drinking Tang, and peeing into tubes just like their ancestors did.” “To them,” Doob said, “that’ll be a back-to-nature experience.”

Highlight (Yellow) | Location 4024

this had led to a kind of renaissance in pedagogy. Free from the constraints of racking up high test scores or getting into colleges, students could learn for learning's sake—which was how it ought to be.

Highlight (Yellow) | Location 4044

“The resources up there are basically infinite. That was true even before the moon blew up. Now it has been busted open like a piñata. All it needs is to be shaped into the right architecture—enclosed habitations that we can fill with air and fertilize with the genetic heritage of the Earth. That's going to take a while, and we'll go through some tough times first. It's going to be tough emotionally when the Hard Rain hits and we have to say goodbye to all that was. And it's going to be tough afterward when the Arkers have to learn how to work together and make hard choices. By far the biggest challenge humanity has ever faced. But we'll survive. We'll use what's up there to build incubators for Our Heritage to live in, to grow, and to improve on what we brought with us. And eventually the day will come when we return. The Hard Rain won't last forever. Oh, it'll last for many lifetimes—as long as human civilization has existed until now. And what it is going to leave behind will be a hot and rocky wasteland. But by that point many generations will have devoted all of their hopes and their creative genius to the problem of remaking the world as well as, or better than, what we see here. We will come back. And that's the real answer, Enrique. Will we survive? Yeah. It'll be touch and go, but we will survive. Will we build space habitats? Absolutely. Small ones at first, big ones later. But that's not the real goal. The real goal will take thousands of years. The real goal is to build Earth again, and build it better.”

Highlight (Yellow) | Location 4073

“Today we're going to talk about what it means to have a swarm of arklets,” he said. “In normal space, like on Earth, we use three numbers to tell where something is. Left-right, forward-back, up-down. The x, y, and z axes from your high school geometry class. Turns out that this doesn't work so well in orbit. Up here we need six numbers to fully specify what orbit an object, such as an arklet, happens to be in. Three for position. But another three for velocity. If you've got two objects that share the same six numbers, they're in the same place. Right now, my six numbers are the same as those of this arklet that I'm floating in, and so we're moving through space together. But if one or more of my numbers changed, you'd see me drifting.”

Highlight (Yellow) | Location 4083

“I can't drift far, confined as I am to the pressure hull. But you can imagine that if I hadn't been able to stop—if I'd been out on a space walk—I might have drifted a long way. And what the science of orbital mechanics tells us is that no two objects in orbit can have the same six numbers, except in the special case I just showed you, where I was inside the hollow arklet so that both of our centers of gravity could coincide. An arklet, or any other object, that is off to the port side of Izzy, or to starboard, or to the zenith or nadir side of it, or forward or aft of it, has different numbers by definition. It's in a different orbit. And so it is going to drift.”

Highlight (Yellow) | Location 4089

If it's in a higher orbit, it'll fall behind. If it's in a lower orbit, it'll race ahead. If it's off to one side or the other, it'll converge, then diverge, on a ninety-three-minute cycle. Only if it's directly forward, or directly aft, will it maintain the same relative position.

Highlight (Yellow) | Location 4092

“The moral of the story? In space, there is no such thing as formation flying. Physics will cause two nearby objects to drift closer together or farther away. If you want to preserve a formation, such as a swarm, you have only two options. Physically connect the arklets together, so that they become one object, or else use the thrusters to correct for the drift.” There was another option, which was to put them in single file, like a train in space, but it didn’t seem very swarmlike and so he left it out of the reckoning for now. Minutes after the video was posted, outraged YouTube commentators would be all over him, pointing out the error and attributing it to dishonesty, incompetence, and/or a conspiracy.

Highlight (Yellow) | Location 4158

“Ninety percent of their diet was prairie dogs. Farmers killed almost all of the prairie dogs and so the population of black-footed ferrets crashed to the point where only seven remained. From that breeding stock, it was necessary to bring them back.” “Wow, only seven . . . so inbreeding must have been an issue?” “We speak of heterozygosity,” she said, “which just means the amount of genetic diversity within a species. In general, it’s a good thing. If you have too little of it, then you start to see the sorts of problems that we associate with inbreeding.” “But if the breeding stock is reduced to only seven . . . then that’s all you have to work with, right?” “Not quite. Well, technically yes, I suppose. But by manipulating some of the genes, we can create heterozygosity artificially. As well as getting rid of some of the genetic defects that would otherwise propagate through the whole population.”

Highlight (Yellow) | Location 4171

we’ll be growing algae as a way to generate oxygen.

Highlight (Yellow) | Location 4179

stuffed birds and boxed beetles and mounted heads of beasts, gathered by Victorian gentleman-collector types in pith helmets, doing their bit for science on the fringes of the empire. Not scientists as we’d define them today but contributors to the scientific ideal. These things overflowed the museums and Clarence acquired them by the lorry-load, especially after Edwina died and couldn’t forbid it. Anyway, I’m that person now, except that the samples are all digital, and they are all on these things.” She tapped a thumb drive that was floating around her neck on a chain. “Or their rad-hard equivalents.”

Highlight (Yellow) | Location 4186

““We can’t send blue whales and sequoias up on the Cloud Ark. And even if we could, we couldn’t keep them alive there. But we can send their DNA, encoded as strings of ones and zeroes.””

Highlight (Yellow) | Location 4192

we can create a strand of DNA from it, beginning with a few simple precursor chemicals. But the amount of human intervention is still ridiculous.”

Highlight (Yellow) and Note | Location 4252

the fact was that an arklet docked to the end of a truss far off to the port or the starboard wing of Izzy was not in a proper orbit at all. Left to its own devices—free, that is, of the constraints imposed, and the forces exerted, by the truss—it would converge on Izzy, cross through Izzy’s orbit, diverge from it, turn around, and converge again, on the same ninety-three-minute cycle that clocked Izzy’s orbits around the Earth. An arklet mounted “above” Izzy on the zenith would want to go “slower” and lag behind; one mounted “below” on the nadir would want to race ahead. To the extent that the truss structure prevented those things from happening—to the extent, in other words, that it succeeded in its basic function of holding all the modules and arklets in a fixed configuration—it was undergoing stress, exerting forces on those arklets to prevent them from doing what they wanted to do. Humans in those arklets would notice themselves drifting and bumping into the walls as their natural trajectories, as ordained by Sir Isaac Newton, were perturbed by the structure of Izzy. The larger Izzy grew, and the more arklets and modules were connected to her, the greater those forces became, and the closer she came to breaking.

Their center of gravity was different despite being connected

Highlight (Yellow) and Note | Location 4263

The space station’s original orbit had been carefully chosen. Any lower, and the thicker air would make the orbit decay too fast. Any higher, and the danger from micrometeoroids would increase.

When earth no longer has atmosphere it could be closer/lower

Highlight (Yellow) | Location 4266

Her orbital altitude of four hundred kilometers was a Goldilocks compromise between “too much orbital decay for Izzy” and “enough orbital decay to sweep the sky of dangerous rocks.”

Highlight (Yellow) | Location 4267

The attachment of Amalthea to Izzy’s forward end a few years ago had changed all of that for the better. Orbital decay was less of a problem because of the rock’s high ballistic coefficient, and micrometeoroids tended to get stopped by the massive nickel-iron cowcatcher.

Highlight (Yellow) | Location 4274

Amalthea couldn’t protect any parts of Izzy that projected beyond its silhouette. Dinah and the rest of the Arjuna Expeditions crew had made some progress in “embiggening” the asteroid, carving out slabs of metal and then elevating them like flaps on an airplane wing to extend the sheltered envelope, but it could only get so big. At some point it had been necessary to draw a line under the expansion and “fix the envelope,” meaning that Izzy took on a definite shape and size. This had occurred on A+1.233. Since then they had found ways to jam more modules inside that envelope, or, where that wasn’t possible, to pack bags and bladders of stored material into gaps.

Highlight (Yellow) | Location 4297

The Cloud Ark passed through a complete day/night cycle every ninety-three minutes. Time was arbitrary in space, so the ISS had long ago settled on Greenwich time, also known as UTC, as a reasonable compromise between Houston and Baikonur. The Cloud Ark had inherited that system, and Day 700 began at midnight at the Royal Observatory in Greenwich, or A+1.335.0 in ark time. About one-third of the population woke up to begin a sixteen-hour shift. Others would wake up at A+1.335.8, or “dot 8,” and at “dot 16.” The system ensured that about two-thirds of the population was awake at any given time. Awake people needed more oxygen and generated more heat than sleeping people, so it put less of a strain on life support systems, and enabled the Cloud Ark to support more people, if the waking and sleeping cycles were staggered. A reason for the popularity of triads was that each of the three component arklets could operate on a different shift, observing its own artificially imposed period of darkness and quiet. In a heptad, the same basic scheme could be used, with two of the arklets asleep at any given time and the one in the middle of the hexagonal frame being “always on.”

Highlight (Yellow) | Location 4332

“Doob? You still with me?” Luisa asked. “I was going to say ‘Earth to Doob, Earth to Doob,’ but it doesn’t apply anymore.”

Highlight (Yellow) | Location 4392

“Politics,” Luisa said. “But we always knew it was going to happen, right?” “Good morning, Doctors,” said a new voice: that of Ivy Xiao, entering the module to begin her own “morning” exercise routine. “Good morning, Doctor,” said Luisa and Doob in unison, though for Doob it was afternoon. “Did I hear the P-word?” “Yes,” Luisa said. “We were just talking about you, honey.” Doob was appalled. But Ivy laughed delightedly.

Highlight (Yellow) | Location 4438

Rhys had been put in charge of building a larger one, concentric with T2, and inevitably called T3. Never one to rest on his laurels, he had invented a completely new system for building it. Unsurprisingly to Dinah, this had consisted of assembling a long loop of high-tech chain and setting it in motion around T2, then adding stuff onto it incrementally. It spun around the same hub at the same RPMs, but because it was bigger, its simulated gravity was a little stronger, about equal to that on the moon. It housed the closest thing the Cloud Ark had to a bridge: a segment of T3 about ten meters long, used by Markus as his headquarters. Attempts had been made to dignify it with names such as “command center,” but at the end of the day it was just an upgraded version of the Banana: a conference room with some television screens and power feeds for tablet computers.

Highlight (Yellow) | Location 4444

Izzy didn’t have a helm. It didn’t have controls as such. No big wheel to turn for steering it through space, no throttle. Just a bewildering assortment of thrusters controlled through a web interface that could be pulled up on any tablet, provided you had the right password. So, the control room, the bridge, the command center could be anywhere. People had ended up calling the room Markus had chosen the Tank.

Highlight (Yellow) | Location 4464

The crawl running across the bottom of the screen was being updated every few minutes by the current BFR, or Bolide Fragmentation Rate, which had started out at zero on A+0 and been climbing ever since then; this was the number that, when it caromed through the bend in its exponential curve, would signal the onset of the White Sky. The networks had been tracking it obsessively. There was an app for it. A bar in Boston had begun offering end-of-the-world drink specials whenever the BFR broke through certain milestones, and the promotion had been copied widely.

Highlight (Yellow) | Location 4476

Selecting the color scheme had consumed resources equivalent to the gross domestic product of a medium-sized country. They had settled on a thin red line down the center, symbolizing the bloodline of the human race, flanked by bands of white, symbolizing starlight, flanked by bands of green, symbolizing the ecosystem that would keep the Arkies alive, flanked by bands of blue, symbolizing water, and, finally, edged by bands of black, symbolizing space.

Highlight (Yellow) | Location 4509

In the middle of all that, the flack in Houston got back to him and asked if he could please relocate to a part of Izzy where zero gravity prevailed, and rerecord the little pep talk. Conspiracy theorists were saying that the Cloud Ark didn't really exist and that it was actually just a bunch of movie sets in the Nevada desert.

Highlight (Yellow) | Location 4526

There were a number of reasons why a big power plant might be needed, but the most important of these was to produce rocket propellant by splitting water into hydrogen and oxygen.

Highlight (Yellow) | Location 4532

“up” in the zenith direction,

Highlight (Yellow) | Location 4534

Certain relics were easier than others to send into space; the Magna Carta had made it up here, but Michelangelo's David was still on the ground. Considerable effort had gone into sealing up heavy relics in “everything-proof vaults” and leaving them on the bottom of the oceans, or in deep mine shafts,

Highlight (Yellow) | Location 4541

he could not humanly ignore the fact that his potential fourth child, the embryo he and Amelia had created, was in there somewhere. Along with tens of thousands of other fertilized embryos waiting to be thawed out and implanted in human wombs.

Highlight (Yellow) | Location 4547

Instead of sending up a church pod, a synagogue pod, a mosque pod, etc., she had provided this one structure, which was something like the interfaith chapel at an airport in that all the different religions had to share it. Internal projectors would display crosses, Stars of David, or what have you on its interior surfaces, depending on what sort of service was happening there at the moment. It had a long, cumbersome, politically correct name, but someone had dubbed it the Woo-Woo Pod

Highlight (Yellow) | Location 4561

size of a coffee table and essentially vaporized. Performing his video surrounded by free-floating, freeze-dried corpses would shut up the “truthers”

Highlight (Yellow) | Location 4615

Somewhere in the middle, between the ocean liner and the blind ox, was the football player pushing the wheelbarrow. It didn’t matter whether “football” for you was soccer or the American sport played by men in helmets. In either case you were meant to envision a player trying to weave a path downfield among defenders. A skilled player could succeed at this when running unencumbered but would fail if obliged to push a wheelbarrow with a boulder in it. The boulder, of course, was Amalthea, and the wheelbarrow was the asteroid mining complex that had been constructed around it. If this analogy were the one closest to the truth, then the wheelbarrow would have to be abandoned.

Highlight (Yellow) and Note | Location 4635

Thus physics drove politics to the extremes of “ditch the rock now” or “keep the rock at all costs.”

And related demand for propellant

Highlight (Yellow) | Location 4665

He was, understandably, quite relaxed at the moment, and it took a while for him to bring his formidable powers of concentration to bear on the matter at hand.

Highlight (Yellow) | Location 4678

“Thank God for Ivy,” Markus said. Since winning command of the Cloud Ark he had never lost an opportunity to praise her—another skill that Doob reckoned must be inculcated into leaders in whatever mysterious Leader Academy churned them out. More likely it was an instinct.

Highlight (Yellow) | Location 4694

she made the most of it, gliding in the liminal zone between dozing and waking

Highlight (Yellow) | Location 4727

It had long since become obvious that he and a number of his mining industry friends had mounted a serious operation to prepare for an extended stay beneath the surface of the Earth.

Highlight (Yellow) | Location 4729

Constructing an underground complex that could sustain itself for thousands of years was an operation of which few, if any, organizations were capable. Most of those were governmental or military. But if any private group could do it, it was Rufus and his network. The sorts of questions he had been asking her for the last two years left nothing to the imagination. To the extent that the experts on Izzy knew anything about long-term sustainability of artificial ecosystems, Rufus now knew it too.

Highlight (Yellow) | Location 4738

all the letters were in the range A through F. Which was a strong hint that this was a number expressed in hexadecimal notation:

Highlight (Yellow) | Location 4744

Some of Dinah's chips were cheap off-the-shelf hardware made for terrestrial use, but she also had some rad-hard ones, which she hoarded in a shielded box in a drawer beneath her workstation. She opened that drawer, pulled out that box, and took out a little green PC board, about the size of a stick of gum, with an assortment of chips mounted to it. Printed in white capital letters directly on the board was its MAC address. And its first half-dozen digits matched those in the transmission coming from the Space Troll.

Highlight (Yellow) | Location 4751

QSB, came the answer back. "Your signals are fading." Then QTX 46, which she guessed meant something like "Will you be available on this frequency forty-six minutes from now?" As anyone on Izzy would understand, this meant "I will call you back when you have orbited around to the other side of the planet." QTX 46, she answered back. "Yes."

Highlight (Yellow) | Location 4784

few sightings on optical telescopes had confirmed that it had transitioned into a heliocentric orbit, which might have happened accidentally or as the planned result of a controlled burn. Assuming it was following its original plan, Ymir should then have made almost two full loops around the sun. Since its orbit was well inside of Earth's—the perihelion was halfway between the orbits of Venus and Mercury—it would have done this in just a little more than a year, grazing the orbit of Greg's Skeleton—Comet Grigg-Skjellerup—a couple of hundred days ago.

Highlight (Yellow) | Location 4796

Time, tide, and comets waited for no man.

Highlight (Yellow) | Location 4814

“My dad used to talk about spark gap transmitters,” Dinah said. “It was a technology they used—” “Back before transistors and vacuum tubes. Yes!” Doob said. Dinah telegraphed down: DOES QET SOUND LIKE AN OLD TIME SPARKY TO YOU? Rufus returned: YES COME TO THINK OF IT “They took some of my robots with them,” Dinah said. “All they would have to do is jot down the MAC addresses on those units’ interface boards, and they’d have sort of a crude proof of identity. As a matter of fact . . .” and she began to pull up some of the records she had made, almost two years ago, of the robots and part numbers issued to Sean and his crew. Within a few minutes she was able to verify that the MAC address that had come in via Morse code a few minutes ago matched one on a robot that had been taken to Ymir.

Highlight (Yellow) | Location 4888

gazing down out her little window, was able to see the terminator—the dividing line between the day and night sides of Earth—creeping over the Thames estuary down below, and lighting up a few tall spires in the London financial district.

Highlight (Yellow) | Location 4927

“Partly that,” Ivy said. “But yesterday they were saying that the whole Venezuelan navy was headed for Kourou to restore order. I’ll bet that navy no longer exists.” “The smaller fireballs? Near the spaceport?” “I’m going to guess fuel-air explosives. They would do almost as much damage as tactical nukes without contaminating the launch site.” Ivy had shrugged loose from Dinah’s embrace and turned around so that her back was to the window. They were now hovering close to each other. Dinah finally got it. “You said that Cal’s boat had popped up. That it was on the surface. That he knew something. You think—” “I know,” Ivy mouthed. Cal had received the order, direct from J.B.F., and he had launched the nuke. He’d probably launched cruise missiles with fuel-air devices as well.

Highlight (Yellow) | Location 5052

“So what is your point?” Markus asked, crisply but politely. “We have no laws. No rights. No constitution. No legal system, no police.” Markus and Tekla looked at each other across the table. It was not a sneaky look, or a guilty look, or a conniving look. But it was a significant look. “It is being worked on,” Markus said. He wasn’t kidding; ever since the Crater Lake Accord had been signed, a whole think tank full of constitutional scholars had been toiling away on it in The Hague, and one of them was now resident up here. “I know it is,” Luisa answered, “and it is very important to me that atrocities such as what we’re seeing on these screens don’t somehow infect that process. This cannot be business as usual.”

Highlight (Yellow) | Location 5062

excused herself from what had become a very strange kaffeeklatsch. Oh well, perhaps it had cured her of some sentimental ideas. She had walked in aspiring to somehow re-create the experience of breakfasting in a sidewalk café in Europe and instead been treated to half an hour of nuclear warfare, mass incineration of protesters, and serious ethical discourse, mixed in with a suddenly keen sexual tension between her and Tekla. Like quite a few other people on the Cloud Ark, she hadn’t had sex since she had come up here.

Highlight (Yellow) and Note | Location 5163

“You figured out a way to make photosynthesis in rice more efficient by transplanting genes from maize. Greenpeace destroyed your research facility in the Philippines but you kept the project alive anyway, in Singapore. Starting shortly after Zero you began developing strains of that rice adapted for cultivation in low-gee hydroponic environments.” “Sprice,” Miguel said, with an ever-so-slight roll of the eyes. The term, a contraction of Space Rice, had been coined by an enthusiastic reporter for the Straits Times and become an unkillable staple of tabloid headlines and Internet comment threads. “Do you understand, Moira, that it cannot grow without some amount of simulated gravity? There has to be an up and a down or the root system cannot develop. In this it is more difficult than algae, which doesn’t care.” “Oh, we’re all going to be eating algae for a long time,” Moira said. “Sprice will come later, after we have constructed more environments that rotate to make gravity. And then, Miguel, then!” “Then what?” Miguel asked. “Sprew.” “Sprew?” “Space brew,” Moira said. “It’s not as good as barley, but you can make beer from rice in a pinch.”

Budweiser utilizes rice.

Highlight (Yellow) | Location 5190

martial arts in zero gee is a new frontier.”

Highlight (Yellow) | Location 5220

enormous sense of shame and guilt comes with it.” “But Amelia doesn’t have Alzheimer’s, she’s—” “Doesn’t matter. Seeing her, talking to her, makes you feel bad. And at some level, your brain wants the thing that makes you feel bad to go away. Simplest reaction in the world. Doesn’t make you a bad person. Doesn’t mean you have to give in to it.”

Highlight (Yellow) | Location 5225

the question of “When?” Predicting it on Day 720, plus or minus a few, had been all well and good back on Day 360. But Day 700 was now approaching its end, and the “plus or minus” thing was seriously bothering him. Lately they’d narrowed it down to “plus or minus three days,” but that was in response to political pressure. It wasn’t a legit scientific move. And it meant something different to scientists. Laypersons understood it as “certainly between 717 and 723.” Scientists would instead say that if you could repeat the experiment of blowing up the moon a large number of times, and keep track of the time-to-White-Sky separately in each case, the numbers would fall into a normal distribution, a bell-shaped curve, with about two-thirds of the instances falling within that range. Which meant that the remainder would fall outside of that range—and some would fall well outside of it.

Highlight (Yellow) | Location 5360

“Door” was a landlubber term for what, on a boat or a spaceship, would be called a hatch. A convention had developed where, in a part of Izzy that had simulated gravity, it was referred to as a door. In the floaty bits, it was called a hatch.

Highlight (Yellow) | Location 5707

“A few minutes ago, Manhattan was struck by a hundred-foot wall of water. I presume that the same is true of most of the East Coast. I was listening to the service from St. Patrick’s Cathedral when it went off the air.”

Highlight (Yellow) | Location 5748

“The Cloud Ark Constitution is now in effect.” Dinah drew breath, knowing what this meant. Markus spelled it out anyway. “This means that all nation-states of Earth, and their governments and constitutions, no longer exist. Their military and civilian chains of command are no more. Oaths you may have taken to them, allegiances you may have held, loyalties you may have felt, citizenships you may have had are now and forever dissolved. The rights granted you by the Cloud Ark Constitution, no more and no less, are your rights. The laws and responsibilities of the Cloud Ark Constitution now bind you. You are citizens of a new nation now, the only nation. Long may it endure.” She keyed: MARKUS IS CALLING IT WHO SAID HE WAS BOSS? Rufus’s transmission was getting scratchy. Dinah wiped her eyes and looked out her window to see Earth encircled by a belt of fire.

Highlight (Yellow) | Location 5799

not a cowboy, and he certainly wasn’t a preacher. He was a miner: a delver, a demolisher, a builder, a businessman. If he lived by a simple code of ethics, it was not an end in itself, but a way to get something done without selling his soul or destroying his reputation. It was a tool to be wielded like a shovel or a stick of dynamite. Tools were for building things; and pride was something you could feel after the fact, when you stood back, looked at what you had built, and passed it on to your children. Dinah could spend the rest of her life living by her word, giving everyone a fair shake, and all of that. Rufus would no doubt approve of all those things. But it was not the charge he had given her. He had told her, though not in so many words, to get busy building a future.

Highlight (Yellow) | Location 6257

Like every other project in the history of the universe, it went slowly at first because all kinds of snags came up.

Highlight (Yellow) | Location 6418

The speed of sound, for example—a.k.a. Mach 1—was three hundred and some meters per second, and most earthbound people would consider it awfully damned fast.

Highlight (Yellow) | Location 6420

A common delta vee benchmark had been the amount needed to get something from an Old Earth launch pad to an orbit like Izzy’s. This was some 7,660 meters per second, or more than twenty-two times the speed of sound:

Highlight (Yellow) | Location 6426

The launch from terra firma to Izzy on Day 68 had required a delta vee of 7,660 m/s according to a naive calculation; but as any old space hand would know, losses due to atmospheric friction and the need to push back against gravity would have elevated the practical number to more like 8,500 or 9,000.

Highlight (Yellow) | Location 6487

Laypersons tended to substitute “fuel” or “gas” for “propellant,” making the obvious analogy to the stuff that had been burned by the engines of cars and airplanes. It wasn’t a bad analogy, but it was incomplete. In addition to fuel, most rocket engines needed some kind of oxygen-rich chemical (ideally, just pure oxygen) with which to burn it. Cars and planes had simply used air. Rockets stored the oxidizer in a separate tank from the fuel until the moment of use. The two chemicals were collectively referred to as “propellant,” and their combined weight and volume tended to dominate space vehicle design

Highlight (Yellow) | Location 6493

A convenient figure for characterizing that was the mass ratio, which was how much the vehicle weighed at the beginning (including the propellant) divided by how much it weighed at the end, when all the tanks had been emptied. If you knew how good the engine was, and how much delta vee you needed, then the mass ratio could be calculated using a simple formula named after the Russian scientist Tsiolkovskii, who was credited with having worked it out. It was an exponential: a fact that explained almost everything about the economics and technology of spaceflight. For if you found yourself on the wrong side of that exponential equation, you were completely screwed.

Highlight (Yellow) | Location 6549

he broke into a string of bad Russian meaning something like I trade places with you now. Vyacheslav responded in equally bad German. Both men were perfectly fluent in English. But they made a private joke of butchering each other’s languages, ostensibly as part of a project to preserve Old Earth’s linguistic heritage. Markus then added, “The rest of you, buckle up.”

Highlight (Yellow) | Location 6628

Unlike normal ships, which carried their propellant in tanks, Ymir was a big chunk of solid propellant—ice—inhabited by a sort of parasitical infestation of equipment whose purpose was to convert that propellant into thrust.

Highlight (Yellow) | Location 6659

The general picture was that Ymir would not be anything like the traditional idea of a spaceship, in the sense of an orderly, symmetrical piece of architecture. It would be more like a flying robotic anthill, constructed out of a natural found object. The robots crawling around on and in it had general instructions as to what they were supposed to be doing, but could make their own judgments from moment to moment to avoid collision with

other robots, or from hour to hour as to when they needed to recharge their batteries. Or that had been the general scheme, anyway.

Highlight (Yellow) | Location 6706

Gamma was high-energy light. It would pass through just about anything. So, bad news, good news: It was hard to shield against the stuff. But most of it passed right through your body without interacting—that is, without doing damage. It made scary noises on the Eenspektor. Betas were free-flying electrons. They were easy to shield against. Good news, bad news: You could stop them with a little bit of water or plastic. But by the same token, if they came into contact with your body they were certain to break something inside of you. Alphas were helium nuclei, four thousand times as massive as betas, moving at relativistic speed. They could not pass quietly through matter any more than cannonballs could, but they did a lot of damage to whatever they hit.

Highlight (Yellow) | Location 6897

Now that fuel fleas were killing people, it was time to learn about them. “A tiny piece of uranium or plutonium that has gotten loose from a ruptured rod. As it throws off alpha particles, it zigs and zags around the room—conservation of momentum. So it hops around like a flea. The point is, it is small and it makes a lot of alpha. It lodged in a diverticulum in his bowel. It burned through his bowel wall and started a bleed that could not stop.”

Highlight (Yellow) | Location 6958

Between being as dead as a doornail and running out of control was a narrow band of normal and healthy power output in which basically all commercial reactor operations happened. The essential problem with Ymir’s reactor was that its moderator—being a naturally occurring substance—was impure and unpredictable. The water that flooded into the chamber for the first dress rehearsal had been melted from ice a few months earlier, around the time of the initial “burn,” and had been sitting in the plumbing system ever since then. There, it had been in contact with rocks and grit that had made it through the augers. It had leached various minerals out of that rock, and become something other than pure water.

Highlight (Yellow) | Location 7016

How hot the air would get, how much it would expand, and how heavy the drag would get were questions of colossal import that, however, simply could not be answered until the Hard Rain actually started. As Doob always put it, the experiment of blowing up the moon had never been attempted before. The most they could do was wait and perform observations. Which was exactly what they had been doing ever since the Hard Rain had begun.

Highlight (Yellow) | Location 7055

Alerted by a schoolteacher standing in the building’s doorway, Camila turned back into the bus. Rather than hitting her full in the face, the shotgun blast raked the left side of her jaw, removing eleven teeth, tearing away much of her cheek, and causing massive structural damage to her jaw. Surgeons in Karachi and, later, London had saved most of the functions of her tongue, rebuilt her mandible from pieces of bone carved from her pelvis,

and fitted her with a set of artificial teeth. After a world tour raising money for girls' education in Afghanistan and the tribal regions of Pakistan, Camila had been granted permanent asylum in Holland. Dutch plastic surgeons, funded by charitable donations from all over the world, had gotten to work repairing the cosmetic damage. This was a long-term project that had been interrupted by Camila's selection as one of the Dutch candidates for the Cloud Ark. No one believed that this was a random outcome of the Casting of Lots. Clearly the Dutch authorities had placed their thumb on the scale and seen to it that she was chosen, as a rebuke to some conservative Muslim countries who had refused to nominate female Arkers unless given assurance that arrangements would be made for them to live in orbital purdah. Camila was well suited to serve that symbolic purpose, since she had not adopted Western ways. She dressed conservatively and wore the head scarf and the face veil. She was coy, however, as to whether the purpose of the face veil was submission to the demands of religion or to hide her disfigurement. She had pulled it down several times to display the scars to television cameras, and when she had dined at the White House she had gone uncovered in the dining room, by prearrangement with her hostess, the president of the United States. Julia's startling arrival in the Cloud Ark had therefore led to a reunion between the forty-four-year-old ex-president and the eighteen-year-old refugee. To call it joyous or even happy would have been wrong given the circumstances. It was a fact of human nature, though, that some people just got on well with each other. This had clearly been the case during that dinner at the White House and was no less true in Camila's abode, Arklet 174, which was where Julia ended up lodging after she had recovered from her eventful flight up and gone through a bit of basic training in how to live in space.

Highlight (Yellow) | Location 7136

Spencer here, after all those years of patient service, was replaced by that hacker boy—"Steve Lake?" Zeke asked. Julia's eyes darted to Camila, who nodded. "Yes," Julia said, "Steve Lake. I guess he's quite clever, but obviously no competition for Spencer." "Are they in competition?" Zeke asked. "In a sense yes, when we Arkies are exposed to the all-seeing eye of SAN, and the GPop is permitted to have some semblance of privacy." "It depends on where you are in the space station," Zeke began, but then trailed off. "I wouldn't know, since I've been permitted to spend very little time there."

Highlight (Yellow) | Location 7146

"To be sure," Zeke said, "and I think that as time goes on those two populations will cease to be thought of as separate groups." "I know that is the official dogma," Julia said, amused. "But most of that social interaction is not going to be through face-to-face visits." "So I'm told. Hard to envision how the populations will merge as long as that is the case." "Most of it is going to be happening through Spacebook and Scape and whatnot," Zeke went on, referring to the Cloud Ark versions of popular Internet communication apps. "At least until—" "Until we all ascend into heaven and live happily ever after as one big friendly Ark," Julia said. "Zeke, you know space operations better than anyone. What is your opinion of the strategy that Markus has been foisting on us? The Big Ride?"

Highlight (Yellow) | Location 7192

THE CREW OF THE ORIGINAL JAMES CAIRD HAD USED CELESTIAL NAVIGATION to find their way across hundreds of leagues of stormy seas to the coast of South Georgia Island. The crew of the New Caird would have to do something similar. It was easier for them. The navigator on James Caird had had no choice but

to await breaks in the ever-present cloud cover and snatch observations when he could, comparing them against a mechanical chronometer that he hoped was still telling true time. New Caird had better timepieces and a better view of the sky. In place of a sextant, they had a device consisting of a wide-angle lens and a high-resolution image sensor that could tell what direction it was aimed in just by comparing what it saw to an astronomical database stored in its memory. So they knew precisely how they were oriented in space, and how that orientation was shifting as the giant shard of ice to which they were attached progressed through the inexorable mathematics of its long ellipse. That, combined with direct measurements of Earth's position, enabled Markus to calculate the parameters of their orbit and to reckon, with precision that grew each time he rechecked the figures, exactly how low they were going to go. Whenever Izzy was on their side of the planet, which was about half the time, they were able to get the latest figures from Doob concerning the expansion of the atmosphere. It was in combining those two sets of figures that pure Newtonian mechanics began to break down. For, in a traditional calculation of a space vehicle's trajectory, one assumed no atmosphere and no extraneous forces resulting from it. But there was now no denying that Ymir would be going low enough to scrape the air. At a minimum, this meant it would experience some drag that would throw it off the course that Sean Probst had laid in. As these things went, drag wasn't that difficult to calculate. Its effect on their course could be estimated. But because the ice shard wasn't a symmetrical body, coming in straight, it was also going to generate some lift.

Highlight (Yellow) | Location 7247

By the time the journey to Izzy was finished, most of the ice would have been fed into the hoppers and blown out the nozzle, leaving a hollowed-out shell with just enough internal structure to hold the reactor in place and maintain some semblance of a nozzle bell. This was not as crazy as it sounded, for two reasons. First, it was what miners had done since time immemorial. They didn't just hollow out mountains, since that would lead to collapse. They sculpted the mountains into structurally sound architectural systems, complete with pillars, arches, and vaults. This was just that, except that the material was ice, and the forces in general were not as large. Secondly, most of the shard's interior was of little consequence from a structural engineering standpoint. There was a reason why airplanes and race cars had been hollow shells—all skin and no bones. Most structural forces were naturally transmitted through the outermost layer of the vehicle, so that was the best place to put the strength. Enough strength on the outside made it possible to leave the inside hollow.

Highlight (Yellow) | Location 7280

Dr. Harris was now wishing that the late Doc Dubois had never opened his big fat mouth. He was in a corner of the Farm that, since the departure of New Caird, had developed into a sort of bullpen where he and Konrad and some of the other orbital mechanics geeks hung out. The Farm had always operated something like a high school cafeteria, with different cliques habitually sitting in certain areas, and now those choices were hardening, becoming a part of Izzy's unwritten procedure manual. Anyway, they had printed out charts and plots representing, in more or less abstract form, everything that they knew about the ongoing development of the lunar debris cloud and what it might mean for the future of the Cloud Ark. The expenditure of paper and printer ink had been somewhat lavish. Two generations from now, if any humans survived, they would look on this heap of documents with some combination of disgust and amazement. Because paper was going to be scarce by then, and they would view its use for such purposes in roughly the same way as Americans of the twenty-first century had viewed the use of sperm whale oil to fuel streetlamps.

Highlight (Yellow) | Location 7297

he was struck by just how young the Arkies were, how different they were demographically from the General Population. It gave him a vaguely troubling sense of being out of touch. It had been decades since he had been young, but he had always been one of the cool kids anyway, with a big following on Facebook and Twitter. Now he was stuck on Izzy and Julia was stuck in an arklet. The two of them were hanging out with completely different populations. GPoppers saw each other all the time and talked face-to-face. Arkies were isolated in their arklets and had to use social media to reach out.

Highlight (Yellow) | Location 7384

“It is a statistical problem,” Doob said. “On about $A+0.7$ it stopped being a Newtonian mechanics problem and turned into statistics. It has been statistics ever since. And it all boils down to the distribution of bolide sizes, and of the orbits in which they are moving, and how those distributions are changing over time—which we can only know from observation and extrapolation. And you know what, Julia? Even if we had perfect knowledge of every single one of those statistical parameters, we still wouldn’t be able to predict the future. Because we have an n of 1. Only one Cloud Ark, only one Izzy to work with. We can’t run this experiment a thousand times to see the range of different outcomes. We can only run it once. The human mind has trouble with situations like that. We see patterns where they don’t exist, we find meaning in randomness.

Highlight (Yellow) | Location 7399

“It would be easier,” Doob said, “if I could figure out what the hell she wanted.” “You’re assuming,” Luisa said, “that she has a plan. I doubt that she does. She is driven to seek power. She finds some way to do that and then backfills a rationalization for it afterward.”

Highlight (Yellow) | Location 7620

“I think she is just being opportunistic. She reaches out to people, draws them into conversations, drops hints, probes for vulnerabilities.” “Is she nuts?” Ivy asked Luisa. “In a sense it doesn’t matter,” Luisa said. “If she’s making trouble, she’s making trouble. Tracing that to a diagnosable psychiatric condition doesn’t really change anything.” “It might change the approach that we take.” “She was narcissistic to begin with,” Luisa said. “This isn’t a formal diagnosis, mind you. But according to what we heard from you and Dinah, her trip up to Izzy was pretty traumatic. She lost her husband and her child and blood was spilled along the way. It doesn’t take a trained professional to guess that she is suffering from some level of PTSD. Connected with that we might expect her to have a dark, paranoid vision of the world. But she may have been that way to begin with.” “She’s cagey,” Ivy said. “As long as all she’s doing is talking to people, there’s not much I can or should do.” “Agreed,” Luisa said. “She is building a political base among the Arkies. If you take some action against her, with your sole pretext being that she’s talking to a lot of people, then you’ve given her just what she wants. But some outreach on your part to the Arkies might be a good idea.” Ivy sighed. “The only answer to politics is more politics,” she said, “and that’s where I’m most useless.” “Deeds, not words,” Zeke said. “That’s what really matters. And when Ymir pulls in, you and Markus will have accomplished something that’s going to make J.B.F. and her clique look puny.”

Highlight (Yellow) | Location 7652

I would suggest that the ability to come and go at will is inextricable from ownership—that's what it would really mean to 'have' an arklet, or a triad, or a heptad." "Well, in that sense we're really all subject to the collective dictates of the swarm," Katherine said. "Parambulator is what decides where we go when." "It truly is one of the most insidious instruments of social control ever devised," Julia said. Katherine looked mildly aghast. "But without it, we have a disaster." "That is what makes it so insidious," Julia said. "One can always justify it by making the safety argument. We will all be slaves of Parambulator until and unless someone decides that some things are more important."

Highlight (Yellow) | Location 7687

In a big organization with a military-style chain of command, you didn't have to be everyone's friend and treasured colleague. Markus understood as much,

Highlight (Yellow) | Location 7782

His feet had sprouted thick white whiskers. It took Dinah a few moments to make sense of the image: he had zip-tied each foot to the back of a Grabb, and the "whiskers" were the protruding ends of the zip ties. It was the kind of improvisation that would have made old-school NASA engineers turn over in their graves, had the Hard Rain not eliminated that possibility. But in the last two years, and particularly the last two weeks, this kind of hillbilly engineering had become routine.

Highlight (Yellow) | Location 7902

"Problem is solved," Tekla said. She was fluent in English and was perfectly capable of saying "The problem is solved," but sometimes dropped the article for effect. Anglophones found this mysterious and impressive. It was also an implicit statement of Russian pride. The language of the Cloud Ark, by default, was English. That was never going to change. But the dialect was going to evolve over time, and Russians could bend it in their direction by finding ways to inject their grammar and vocabulary into everyday speech. "Burn is complete," she went on. "But the ship is still tumbling out of control!" said the American boy who so fancied his own intelligence. "Slow tumble," Tekla said. "Not problem. Plenty of time to fix now that perigee is raised." "Fix it how?! Markus demolished three of the external thruster packages by ramming them! Who does that? Anyway, there are only two of them left. It is a basic reality of physics that you can't control a three-axis tumble using only two thrusters!" "Thank you for explaining basic reality," Tekla said. "Tumble can be eliminated by making scarfed nozzle." This silenced them for a few moments. One of Julia's followers—Jianyu, a Chinese Arkie, very passionate about going to Mars—looked like he understood it. Tekla nodded in his direction. "This man will explain later. My time here is limited."

Highlight (Yellow) | Location 7924

The statement was met with perfect, deadpan silence. Tekla just waited. Soon the temptation to boast would get the better of them. Her training in tradecraft—in how to be a spy—had not been all that extensive. A few basic courses, some assigned reading. The reason was simple: She was too conspicuous to be useful as a spy. Too

similar to the Hollywood profile. Real spies went unnoticed. So they had kicked her out of the program and put her to work in roles, such as being an Olympic athlete, where her conspicuousness was an asset. But she had picked up a few general precepts. And she knew that this one thing—the urge to boast of one’s accomplishments—had betrayed more secrets and destroyed more careers than anything else. She looked at Grindstaff. Unlike most people, who soon broke eye contact, he looked right back at her, grinning. “Unusual,” Tekla said, “for one of your background.” “Sources and methods,” he said.

Highlight (Yellow) | Location 7950

Tekla favored the circle of hangers-on and admirers with a long, slow panoramic look, then actually rolled her eyes. “Everyone out!” Julia commanded. “You too, Spencer. Just Tekla and me.” “Your tradecraft is of low quality,” Tekla said, when all the others had dispersed through the hamster tubes to the other arklets in Julia’s heptad. “I know,” Julia said. “It is so difficult rebuilding an intelligence community from scratch.

Highlight (Yellow) | Location 7958

Julia narrowed her eyes. “What have I missed that is not so obvious?” “You should not trust Zeke Petersen with further information,” Tekla said. “Unless you wish to plant false intelligence, in which case he will be an effective channel.” Ivy and Zeke and Tekla had discussed it beforehand, and Zeke had cheerfully volunteered to be given up by Tekla as a supposed turncoat. It made little difference to him personally. And it would go a long way to cementing the idea, in Julia’s mind, of Tekla as a master double agent. By Cold War standards it was an obvious and amateurish gambit, but this was not the Cold War. This was a small town of fifteen hundred people with a former mayor who was trying to stir up trouble. Julia narrowed her eyes and nodded slowly. She was fascinated. “I had wondered about him,” she said. “He seemed like he was just playing along. Just being polite.” “Not a problem with Tekla,” Tekla said.

Highlight (Yellow) | Location 9491

“We don’t need sperm,” Moira said. “We don’t need sperm to get pregnant! This is news to me,” Aïda said, with a sharp laugh. Moira went on coolly. “There is a process known as parthenogenesis, literally virgin birth, by which a uniparental embryo can be created out of a normal egg. It’s been done with animals. The only reason no one ever did it with humans is because it seemed ethically dodgy, as well as completely unnecessary given the willingness of men to impregnate women every chance they got.”

Highlight (Yellow) | Location 9511

this version of parthenogenesis, which is, again, that all offspring are the same. Exact copies. To get some genetic diversity, we need to use something called automictic parthenogenesis. Look, it’s a long story, but the point is that in normal sexual reproduction there is crossing over of chromosomes during meiosis. It’s a form of natural recombination of DNA. It’s what causes your children to look sort of like you, but not exactly like you. In the form of parthenogenesis that I am proposing to use, there would be that crossing over. An element of randomness.”

Highlight (Yellow) | Location 9516

“And both boys and girls?” Dinah asked. “That’s harder,” Moira admitted. “Synthesizing a Y chromosome is no joke. My prediction is that the first set of babies—perhaps the first few sets of them—will all be female. Because we simply need to get the population up. During that time I can be working on the Y chromosome problem. Later on, I hope that some little boys will result.”

Highlight (Yellow) | Location 9521

“So they’ll be quite similar to us genetically.” “If I do nothing about it,” Moira said, “they’ll be like sisters. Perhaps even more similar than that implies. But there are a few tricks that I can use to create a wider range of genotypes out of the same source material. Perhaps they’ll be more like cousins. I don’t know, it’s never been tried.” “Are we talking about the inbreeding problem? It sounds like it,” Dinah said. “Loss of heterozygosity. Yes.

Highlight (Yellow) | Location 9533

“Everyone has some genetic defects. When you are breeding more or less randomly within a large population, there’s a tendency for those errors to be swamped by the law of averages. Everything sort of works out. But when two people sharing the same defect mate, their offspring is likely to have that defect as well, and over time we see the usual unpleasantness that we all associate in our minds with inbreeding.” “So,” Luisa said, “if we follow the plan you have laid out, and begin, a few years down the road, with seven groups of what amount to siblings or cousins—” “It’s not enough heterozygosity, to answer your question,”

Highlight (Yellow) | Location 9540

“As it happens, Old Earth compiled vast databases on such things before its destruction. All of which are in there now.” She gestured in the direction of her lab. “We have a very good idea which defects, on which chromosomes, are responsible for alpha-thalassemia. If you supply me with an ovum, I can find those defects and I can fix them before we begin parthenogenesis. Your offspring will be free of that defect. Barring some random future mutation, it’ll never return.”

Highlight (Yellow) | Location 9932

In truth, there was no aspect of this flight that could not have been managed by a robot. Robot gliders were at this moment operating all over New Earth. But she was leery of letting her own powers dwindle by delegating them to machines, and so she liked to fly the glider at least part of the time. The algorithms worked, but they wouldn’t get better unless humans gardened them; and to do that, you had to fly.

Highlight (Yellow) | Location 9979

It was much too large a volume to be pressurized. What little air it had scooped up during its dip into the atmosphere rapidly leaked out. So Kath Two was effectively in outer space now. Knowing this, the fabric of the suit had contracted against her skin to supply the back pressure that was no longer provided by the atmosphere.

It was porous, and so the only thing really between her skin and the void was Space Grease. The combined effects of that and the nat mesh fooled her skin and muscles into believing that they were under a nice thick blanket of air—the way humans were meant to live. The only part of the outfit that was pressurized like an old-fashioned space suit was the helmet.

Highlight (Yellow) | Location 9999

Something strange caught her eye, and she went to the effort of rotating her head slightly just to verify it was real: the third pilot was actually walking upright. He was trudging along with short deliberate strides, carefully judging his balance and the loads he was placing on his joints, while somehow managing to keep enough blood in his brain to remain conscious. Kath Two could never have stood up, let alone walked, under three gees. The same was true of most of her race. This man, however, was a Teklan. That was obvious from his size, as well as his coloration and the shape of his head, which were visible through his helmet. It was hinted at by his musculature and by the style of the suit he was wearing—heavier, partly armored, slung about with load-bearing straps made to support various burdens. His scabbards, holsters, and bandoliers were vacant. Even without any of those clues, however, she could have guessed his race from the fact that he had chosen to perform the feat of walking when he could have more safely and more easily crawled. Had it not been for the racial bond that joined Moirans and Teklans, Kath Two might have rolled her eyes and muttered a joke about it. Teklans didn't need blood in their brains to keep dutifully trudging forward. Something along those lines. But that kind of stereotyping could just as well have been turned back on her. The Teklan had piloted his vehicle into the hanger under power. The thing had engines. Why not use engines if you belonged to a civilization that knew how to make them? Kath Two, on the other hand, had reached the same destination in an unpowered glider, using skill and wit to draw energy from the atmosphere. She could have turned pilot's duties over to an algorithm anytime she wanted. Instead, she had chosen to do most of it herself. In its own way, this was no less an act of pointless bravado than what the Teklan pilot was doing right now. She had been testing and sharpening a set of skills that was important to her. Mutatis mutandis, this Teklan was doing the same.

Highlight (Yellow) | Location 10179

he had been one of many scientists who had rebelled against a tradition of TerReForm thought that had passed as gospel for hundreds of years previously. Thanks in part to the young Turks' propagandizing, this older approach had become hidebound and stereotyped as the TOT, or Take Our Time, school. The premise of TOT was that ecosystems—which on Old Earth had evolved over hundreds of millions of years—would have to be rebuilt slowly, through a sort of handcrafting process. Which was fine, since living in habitats was safer and more comfortable anyway than the unpredictable surface of a planet. The human races could enjoy thousands of years of safe, secure habitat life while slowly re-creating ecosystems down below that would resemble those of Old Earth.

Highlight (Yellow) | Location 10185

The planet would become a sort of ecological preserve. Africa, whose outlines were still vaguely recognizable, though heavily reshaped by the Hard Rain, would have giraffes and lions sequenced from the ones and zeroes dating all the way back to the thumb drive around Eve Moira's neck.

Highlight (Yellow) | Location 10213

It had thus paved the way for the Great Seeding, which was responsible for the trees that Kath Two had been flying over this morning. In subsequent decades, larger and larger animals had been set loose on the surface as part of a planned program to jump-start whole ecosystems. Some of those—the ones Kath Two had been worried about this morning—were canids. When Rhys said that they were “going epi,” he meant that they were passing through some kind of epigenetic shift.

Highlight (Yellow) | Location 10221

A small minority of those DNA sequences had clearly defined functions. The remainder seemed to do nothing, and so were dismissed as “junk DNA.” But that picture had changed during the first part of the twenty-first century, as more sophisticated analysis had revealed that much of that so-called junk actually performed important roles in the functioning of cells by regulating the expression of genes. Even simple organisms, it turned out, possessed many genes that were suppressed, or silenced altogether, by such mechanisms. The central promise of genomics—that by knowing an organism’s genome, scientists could know the organism—had fallen far short as it had become obvious that the phenotype (the actual creature that met the biologist’s eye, with all of its observable traits and behaviors) was a function not only of its genotype (its DNA sequences) but also of countless nanodecisions being made from moment to moment within the organism’s cells by the regulatory mechanisms that determined which genes to express and which to silence. Those regulatory mechanisms were of several types, and many were unfathomably complex. Had it not been for the sudden intervention of the Agent, the biologists of Old Earth would have devoted at least the remaining decades of the century to cataloging these mechanisms and understanding their effects—a then-new science called epigenetics.

Highlight (Yellow) | Location 10318

The Eye observed whatever time was local on the part of Earth lying directly below it. Currently, it was about eight in the morning there. She could look forward to some serious jet lag—another term from the pre-Zero era that had become embedded in the language despite the obsolescence of its literal meaning.

Highlight (Yellow) | Location 10405

On the social front it was a question of Amistics, which was a term that had been coined ages ago by a Moiran anthropologist to talk about the choices that different cultures made as to which technologies they would, and would not, make part of their lives. The word went all the way back to the Amish people of pre-Zero America, who had chosen to use certain modern technologies, such as roller skates, but not others, such as internal combustion engines. All cultures did this, frequently without being consciously aware that they had made collective choices.

Highlight (Yellow) | Location 10466

The same service could have been provided by a bubble of glass on the flivver’s hull, but it wouldn’t have been as good. It would have exposed the user’s head to cosmic radiation, and the contrasty light would have made it difficult to see certain things. The varp, on the other hand, played games with the light’s dynamic range so that

bright things were less so while dim things were bright enough to see; it gave everything a luminous warm quality that did not exist in reality. It was so far superior to looking at the world directly that many space suits eschewed transparencies altogether and just encased the wearer's head in a radiation-shielded dome with a varp on the inside. She was now "looking" at an enhanced view of the universe from their current location, which was just inside, but rapidly approaching, the habitat ring.

Highlight (Yellow) | Location 10526

The funny thing was that the entire procedure, from Kath Two's landing her glider in the hanger, to the bolo release, to the just-concluded interaction with the whip, would have looked graceful, even gentle, when viewed from a distance. In order to understand the sheer intensity of the acceleration and the jostling, you had to live through it. Trying to think about something other than her stomach, she took a good look at the Eye. Right now it was encircling Akureyri, a big (population 1.1 million), newish (eighty-four years old) habitat constructed in the style known informally as the double barrel, historically known as the O'Neill Island Three type. It was two large cylinders, parallel to each other, rotating in opposite directions. Each was enshrouded in complexes of mirrors and other infrastructure. The mirrors were aimed at the sun, whose light was bounced through windowed strips on the cylinder walls

Highlight (Yellow) | Location 10880

it was possible to look at actual smartphones and tablets and laptops that had been manufactured on Old Earth. They did not work anymore, but their technical capabilities were described on little placards. And they were impressive compared to what Kath Two and other modern people carried around in their pockets. This ran contrary to most people's intuition, since in other areas the achievements of the modern world—the habitat ring, the Eye, and all the rest—were so vastly greater than what the people of Old Earth had ever accomplished. It boiled down to Amistics. In the decades before Zero, the Old Earthers had focused their intelligence on the small and the soft, not the big and the hard, and built a civilization that was puny and crumbling where physical infrastructure was concerned, but astonishingly sophisticated when it came to networked communications and software. The density with which they'd been able to pack transistors onto chips still had not been matched by any fabrication plant now in existence. Their devices could hold more data than anything you could buy today. Their ability to communicate through all sorts of wireless schemes was only now being matched—and that only in densely populated, affluent places like the Great Chain.

Highlight (Yellow) | Location 10914

The end result, for a young woman in a bookstall above a tube station on the Great Chain, was that she was dwelling in habitats, and being moved around by machines, far beyond the capabilities of Old Earth. She was being served and looked after by robots that were smarter and more robust than their ancestors—the Grabbs and so on that Eve Dinah had programmed on Izzy. And yet the information storage capacity of her tablet, and its ability to connect, were still limited enough that it made sense for her to download books over a cable while that was easy, and to make room for them in the tablet's storage chips by deleting things she had already read.

Highlight (Yellow) | Location 10969

Cables broke all the time, because they'd been hit by bolides or simply because they had "aged out," and so if you squinted your eyes and looked closely enough at the tether, you could see that it was alive with robots. Some of these were the size of buildings, and clambered up and down the largest cables simply to act as mother ships for swarms of smaller robots that would actually effect the repairs. This had been going on, to a greater or lesser extent, for many centuries.

Highlight (Yellow) | Location 11011

"All duty is inconvenient to a greater or lesser degree, or it would not be duty."

Highlight (Yellow) | Location 11115

Limitations on building height led people to dig down into the underlying metal, rather than building upward, and so most of the city's square footage was hidden. The buildings were like icebergs, larger below than above.

Highlight (Yellow) | Location 11215

Consciously or not, the Seven Eves sorted themselves into Four, Two, and One. The Four were Dinah, Ivy, Tekla, and Moira. The Two were Camila and Aïda. Arithmetic suggested that the descendants of the Four would be about as numerous as those of the Two.

Highlight (Yellow) | Location 11266

Julia, the One, looked to make the best of a bad situation by endowing her offspring with qualities that would make them useful and important in spite of much smaller numbers. She had already expressed, during the Council of the Seven Eves, the idea that there was real value in the ability to envision possible futures. And she had linked it to leadership, or, failing that, to the ability to give useful advice to leaders. When that trait ran out of control and sought dark paths it led to depression, paranoia, and other forms of mental illness. The challenge then was to find a way of combining that trait with a more positive mentality. Julia's research—and she did a lot of research—therefore tended to center on the history of sages, seers, ecstasies, shamans, artists, depressives, and paranoiacs throughout history,

Highlight (Yellow) | Location 11296

This had occupied much of the first half of the First Millennium and led to a condition of overcrowding so severe that it had made obligatory the formation of separate colonies away from Cradle. For there were other places, perhaps not quite so favored as Cleft, but still well suited for the building of new habitats. They had reached the point, by then, of being able to construct new machines for moving about in space. It was time. Or so insisted the descendants of the Four, who sensed that conditions had become inimical to them in the crowded precincts of Cradle. Camila had been frank about her strategy of making new humans well suited to life in confined spaces. She had succeeded in doing so.

Highlight (Yellow) | Location 11513

So, she inferred he was making reference to some kind of back room or snuggery that she would never be able to find on her own. “Came up the back way,” he added. “There’s a back way?” “There’s always a back way.” “Doc?” “Showed up half an hour ago.” For the most important living architect of the TerReForm to walk into the front door of a crowded bar on Capitol Hill would be to create all manner of unnecessary distractions. Doc would be recognized.

Highlight (Yellow) | Location 11566

like many Europeans, she was part Neanderthal.

Highlight (Yellow) | Location 11570

Aïda had become fascinated by the idea that Africans carried more genetic diversity within their genomes than non-Africans, as a simple result of the fact that humanity had originated on that continent and spread outward. Non-African races had been founded by isolated groups of adventurers. Breeding among themselves, they had created gene pools that were necessarily limited to what they had brought with them: only a subset of what was to be found in Africa. This idea had been used to explain, for example, why Africa contained both the tallest and the most diminutive people in the world, and why so many top athletes were African. It wasn’t because they were naturally better athletes but because the bell-shaped curve of random genetic variation was wider. For every African who was a great athlete there was presumably another who was miserably uncoordinated, but no one paid any notice to the latter. Whether or not this was a valid theory, the fact was that Aïda swallowed it hook, line, and sinker and used it to inform her genetic strategy in the Great Game. And to the extent that the Four bothered to develop counterstrategies, they had to take it into account. The very existence of Moirans, as a race, was a result. Rather than try to follow all of Aïda’s machinations in detail, base pair for base pair, Eve Moira had chosen to tinker with those aspects of the genome that controlled epigenetics, making her children into Swiss Army knives.

Highlight (Yellow) | Location 11771

Ty’s staff, well accustomed to these surges, began to open up back rooms. Extra cooks came up through one of the back entrances and began to make use of mise en place that had been prepped the night before. Everything, in other words, ran smoothly. Which was how Ty liked it. The ability of the Crow’s Nest to accommodate a socket surge with no intervention from Ty, other than polishing a glass, was, in a sense, his life’s work. He had done every job it was possible to do in this place, from floor mopper on upward, and learned over time to select and delegate the work to others who could do it better. He had advanced, in other words, to higher levels of mental activity while always doing enough of the floor mopping and glass polishing to remain in physical contact with the business of the bar and in human contact with the staff. His real job—the job that the Owners paid him for—was to be an observer of the human condition as it was so richly displayed from day to day within these walls. He was also a judicious manipulator of the human condition in the sense of occasionally throwing people out, telling others to settle down in a manner so smooth and humorous that they didn’t know they’d been told, and making certain others feel welcome when they seemed ill at ease. All of that was as fundamental to the operation of a bar as mopping the floor.

Highlight (Yellow) | Location 11847

awaiting the moment when Doc, or some successor of his, would cut the ribbon on New Earth and allow settlers to flood in, had complicated feelings about Sooners and Indigens. On the one hand they were viewed as sharp operators. Tricksters. At the same time they were isolated bumpkins. Ty had learned early how to play both sides of the image. A stranger from the ring who took you for a wide-eyed rube would spill a lot of information before he came to understand the truth, and one who expected you to play tricks on him would let down his guard at the first show of honesty and plain dealings.

Highlight (Yellow) | Location 11852

IF YOU TOOK A LARGE NUMBER OF FLYNKs—FLYING, AUTONOMOUS chain links—and joined them together into a long chain, and connected its ends to make it into a continuous loop, and then got the whole loop moving through the air like a train composed of little airplanes, each using its stubby winglets to generate its share of the lift, then you had a thing known as an “aitrain,” pronounced the way a resident of Old New York would have said “A train.” The concept was old enough that its etymology had been obscured by time.

Highlight (Yellow) | Location 11890

These, like all the other vanes on all the other thousands of flynk in the chain, effected tiny adjustments to keep the whip trimmed in just the right configuration. The result, a few moments later, was that the glider came snapping over the top just as its connection to the last flynk was severed. In a few seconds it had been hauled two thousand meters straight up and let go with a velocity of a few hundred kilometers per hour. Meanwhile, every other flynk in the chain had decoupled itself fore and aft, causing the entire chain to disintegrate into a linear cloud of identical fragments, each headed in a different direction. Each flynk, sensing that it was aloft and alone, automatically deployed large tail vanes that turned it from a bullet into a badminton shuttlecock. The flynk rapidly slowed down to their terminal velocity, turned nose down, and began to fall toward the ground. A slight canting of the vanes caused them to begin spinning like maple seeds, further slowing their descent, and in this manner the entire swarm began to descend in the direction of an empty lot adjacent to the flynk barn.

Highlight (Yellow) | Location 11901

The first half hour of the flight was a little unsettling to Ty’s stomach as Kath Two made a few sudden maneuvers, perhaps because she had sensed good air in one direction or bad in another. People who were accustomed to flight in powered aircraft frequently had trouble adjusting to the unpredictability of gliders, but Ty, who had done it before, understood that Kath Two was just hunting for the right way to inject them into the mountain wave hovering invisibly in the upper atmosphere above the Andes crest. He knew she had found it when the juking and jiving stopped and the back of the seat pressed him forward with palpable acceleration. They were in level, steady flight now, proceeding north at something like three hundred kilometers per hour. Kath Two’s task henceforth would be to look far into their future with her lidar-enhanced sensorium and make small adjustments needed to dodge pockets of rough air.

Highlight (Yellow) | Location 11942

When Ty pointed that out, Bard said yes while shaking his head no, as if to say, But of course, what you are saying is obvious.

Highlight (Yellow) | Location 11997

The lack of a moon meant that New Earth's tides were caused entirely by the gravity of the sun, which made them weaker and more closely synchronized with the cycle of night and day. Because tidal zones were thought to have disproportionate importance to the ecosystems of land and sea alike, much TerReForm brainpower had been focused on them, and the low-lying banks of wave-washed rubble around Magdalena had become spawning grounds, not just for fish and birds and crustaceans, but for researchers with advanced degrees.

Highlight (Yellow) | Location 12050

As a younger man he had tolerated those as part of the cost of doing business—a sort of tax that all people must pay in order to take part in civilized society. When he had turned one hundred, he had decided to stop paying that tax. Henceforth he would engage only in conversations that really interested him—which, with a few exceptions for close friends and family members, meant conversations with a purpose.

Highlight (Yellow) | Location 12189

There was nothing wrong with that. So people on the ring kept saying, because it was the polite thing to say. Nothing wrong with breeds. But the truth of it was that breeds, like weeds, tended to be found in disturbed areas. A sprinkling was nice, especially in sophisticated places like Chainhattan, but seeing a lot of them in one community was a sign that everyone on the ring knew how to read, even if they knew it was impolite to say what they were thinking. The behaviors that these Moirans had invented around everyday things like the sunrise, mealtime, and the interpretation of dreams had a ritualistic quality that to Ariane was obviously fascinating and to Kath Two was a little mortifying. For the first time in her life she was feeling the stirrings of what was called Old Racism: the survival into modern times of racial attitudes, or reenactments thereof, that had existed on Old Earth, had been altogether snuffed out, and were known only because documentation thereof had survived. On a certain kind of diseased mind they exerted the same magnetic pull as they had pre-Zero, and so among a population of millions on the ring you might find one person who'd spent too much time delving into a five-thousand-year-old web archive and become infected with ideas about pre-Zero blacks that he fancied were applicable to Moirans, and so on. It was purely an intellectual curiosity and not at all a factor in real people's lives: a thing Kath Two had heard of, like rabies or Watergate, and she was fascinated to find it stirring in her own mind here of all places. But that was only a passing notion.

Highlight (Yellow) | Location 12204

A weird culture, utterly nontransferable to the ring, had formed at this one. If it had happened here, something like it must have happened at others. How many? Was New Earth infested with bizarre cultural outliers centered on TerReForm installations?

Highlight (Yellow) | Location 12623

“Any well-informed Spacer knows the context,” Memmie returned, sweeping her free hand up across the sky. “It is only a certain type of mind that scorns what is known by all and treats secrets as jewels.” After this Ariane seemed to feel that further conversation would not be to her advantage. It was only the ten millionth such conversation that had taken place during the long, fraught, weirdly personal relationship between Julians and Camites, so Ariane knew her cue to shut up and look aggrieved.

Highlight (Yellow) | Location 12686

“I may look five thousand years old,” he began, “but I am in truth a mere descendant of those you style cowards. Though I daresay you would take a more charitable view if you knew of the deeds that they performed during their long exodus. Do I have the honor of addressing one whose ancestor was Rufus MacQuarie?” “We are all of that lineage,” the old man boomed. “Then I think I have something that belongs to you,” Doc said. Moving deliberately, he pulled the stick out of its purchase on the ground and hefted it up until it lay horizontally across both of his outstretched palms. “Please accept my apologies for having borrowed it without your remit.” Had the Spacers been able to watch all of the Diggers’ reactions at once, it would have yielded a bonanza of intelligence about the workings of their minds and of their society.

Highlight (Yellow) | Location 12694

The young males seemed divided between a more vindictive group who wanted the stick of Srap Tasmaner confiscated immediately, and others of a more chivalrous turn of mind. The group at large comprised a minority who were indignant about Doc’s expropriation of their shovel handle, but—more significantly—a majority who felt ashamed at the idea that they would take away an old man’s walking stick. What those two groups had in common was that they took what Doc had said at face value. A smaller inner circle—the old man, the younger leader, and a woman of intermediate age who had stepped forward to confer with them—had the wit to understand that Doc was playing to the crowd, and not actually trying to initiate a conversation about the ownership of a piece of wood. In other words, the Diggers were, as a whole, reacting much as any other group of humans might have done.

Highlight (Yellow) | Location 12714

The younger man meanwhile launched himself at Ty. It was not clear whether his intention was to inflict damage or merely to restrain him while Remembrance was sacrificed. In any case his foot dislodged a stone as he made his move, creating a sound that gave Ty a bit of warning. He was able to turn so that the attacker rolled around him rather than striking him dead-on. The Digger’s downhill movement became a disadvantage, overbalancing him. Hooking the man’s ankle and kicking upward, Ty made him fall hard on his face. Both of his feet were projecting up in the air, soles of his moccasins exposed to the sky. Ty hooked a leg through the crook of a knee and then dropped his weight, bending the foot forward; the man’s heel would have gone all the way to his buttock had Ty’s lower leg not been interposed behind the back of his thigh and his upper calf muscle, where it threatened to rip the man’s knee apart. The man scrabbled at the ground and so Ty put more weight on it, the stink of the old moccasin in his nostrils, and felt a preliminary pop inside of the joint. The man screamed and stopped struggling.

Highlight (Yellow) | Location 12870

“Where did you learn how to fight like that?” Einstein asked him. “Are you part Teklan or something?” “Fighting isn’t about knowing how,” Ty said. “It’s about deciding to.” “Well, I was frozen, man.” “Look, these are times when the decisions that our Eves made five thousand years ago control our actions to a degree that renders us basically helpless. You were meant to stand back and observe and analyze.” “And you were made to be a hero,” Einstein said. “A hero would have saved Mimmie.”

Highlight (Yellow) | Location 12879

There were a lot of kids under ten years old, but few teenagers. Half of the women looked pregnant. “Play your role, Einstein. You’re the Ivyn in our group now that Doc’s gone. What do you see?” Einstein seemed reluctant to speak, so Ty prompted him: “I see a population explosion.” Einstein snapped into focus and nodded his head. “You’ve never heard of these people,” Ty went on, “even though your RIZ is just on the other side of these mountains and your people patrol up here all the time.” “Rufus MacQuarie’s mine was far to the north,” Einstein offered. “These people must have just come out into the open recently.” “Look for the oldest child down there and that probably tells you the date.” “But the atmosphere’s been breathable for three hundred years! Why would they wait until now?” Ty nodded toward the center of the Digger camp: the big bed of coals, the meat roasting over it. “Food?” Einstein said. “Food and fuel,” Ty confirmed. “They’ve been down in their hole living on God only knows what—cave tofu or something—since the beginning of the Hard Rain. Every so often maybe they would sample the air outside. When it became breathable they probably went out and had a look around. But it was still a wasteland, not capable of supporting life. It’s only in the last few years that TerReForm has seeded that part of Beringia with animals big enough to be worth the effort expended in hunting them. That was the starting gun—the signal for them to come out.” “And to begin having kids as fast as they could, apparently.” “Apparently.”

Highlight (Yellow) | Location 12895

Now, Einstein, what does that tell you about gender roles?” “Well, to begin with, they don’t have an Eve, they have an Adam—Rufus—so it could easily be more pat—patree—” “Patriarchal.” “Thanks. And then if all the women are expected to make lots of babies—” “That tells you something,” Ty said.

Highlight (Yellow) | Location 12900

over time you can see the Eye moving back and forth across it and you can see new habitats lighting up as they are built. You can see bolos coming in low across the sky and TerReForm aircraft flying right overhead and showers of ONANs coming down straight from the ring. And you’re no ignorant savage. Your folk have maintained a reasonably advanced engineering culture. Those compound bows. That lump of explosive. So you wouldn’t have interpreted all of that as gods or angels or any of that.” “They’ve known,” Einstein said. “From the first—” “For centuries,” Ty said, nodding. “As long as they could breathe outside.” “For that long, they have known that billions of humans were living in the sky,” Einstein said. “But they didn’t make any attempt to signal.” “More than that. They hid from us!” Ty said. “Efforts were made, you know, to find the MacQuarie mine several decades ago. These people must have made some kind of decision that they didn’t want to be found.”

Highlight (Yellow) | Location 12914

“If I’m squatting in my mine shaft eating cave tofu when I know perfectly well that there are lots of humans up in geosync living in better conditions, then I need some kind of powerful incentive to stay in my cave. To conceal my presence.” “Some kind of dukh or I dee—” “Ideology,” Ty said with a nod. “I should have seen all this. God damn me for not seeing it a few minutes quicker.” “Seeing what?” “That only a mind virus, a shared hallucination, could explain the suddenness of their appearance aboveground.”

Highlight (Yellow) | Location 12941

She was the worst nightmare of any Julian trying to live an honest life in Blue: someone ostensibly Blue who turns out to be a Red mole. How long had she been insinuating herself into intelligence, working her way up the ranks? Or had she only just decided to switch sides? In either case, she was up in the Red part of the ring now with the woman she had abducted.

Highlight (Yellow) | Location 12954

“I am Donno,” announced the graybeard. “To me you may speak, but no others save the Psych here.” Or at least that is what Ty thought he heard. “I am Tyuratam Lake,” Ty said. “And this is Einstein. The woman there is Kath Two; she is unlikely to join the conversation.” “Tyuratam,” said the Psych in a husky voice, “a city in Central Asia, close to the Soviet space launch facility of Baikonur in Kazakhstan. Einstein, a theoretical physicist of the early twentieth century, before Zero.”

Highlight (Yellow) | Location 12960

“When Kath Two awakens, you will tell her the rule I have just proclaimed,” Donno said, “and see to it that she abides by it.” “I will tell her the rule,” Ty said, “and she will keep her own counsel as to abiding by it. Over her I wield no authority. It is not how our society is organized.” Donno looked as though he didn’t believe a word of what Ty had just said. “You are Dinan.” So, they knew about the Seven Eves. How had they come by that knowledge? Abducting stragglers, interrogating them? Or had they been in covert contact with some Spacer? “Yes,” Ty said. “You are the leader of the group.”

Highlight (Yellow) | Location 12971

“The other mutant—a Julian?” “Yes.” “She attacked you with your own weapon. You were surprised.” “Indeed I was, Donno.” “She betrayed you?” “Yes.” “Is she of the western people?” To Donno, that would suggest the Spacers living in the part of Beringia west of 166 Thirty. “Red is what we call them.” Donno nodded as if he’d heard it before. “You are Blue, then.” “Yes, we are Blue. We avoid using Thors.” “Thor: a Germanic deity of immense strength, associated with lightning, armed with a hammer,” the Psych said. “Is your name short for Encyclopedia?” Einstein asked her. Donno threw Einstein a killing look. Einstein was oblivious to it; he was looking at the girl with fascination and then some. “Yes,” she answered before Donno could stay her by raising his hand.

Highlight (Yellow) | Location 12988

“He is Ivyn,” Donno said, nodding at Einstein. It wasn’t a question. Then, his initial flash of anger having cooled, he took a more careful look at the kid from the RIZ. “His eyelids look that way because of epicanthic folds,” said the Cyc, who had been conducting an unnecessarily close inspection of the Ivyn’s face.

Highlight (Yellow) | Location 13024

Ty corrected him. “We were a Seven, yes. That is a grouping that we create on special occasions, when we need a formal delegation.” What he said next was guesswork, but he needn’t worry about being contradicted at this point. “The old Ivyn who is now dead—Doc, we called him—suspected that you were down here. He came down to investigate, and he did so as part of a Seven. Befitting its importance.”

Highlight (Yellow) and Note | Location 13037

“Your warriors are taking ours,” Donno complained. “As one who knows something of warriors,” Ty said, “you can well imagine how this all looks to them.” He closed his hand around the chain and gave it a little shake. Again, it was the wrong thing to say. The mere suggestion that it might be possible to look at a thing from more than one point of view was infuriating to these people. Ty needed to get that fact through his head.

In a Patriarchal society having a single anlitcal thet overlyh single point of view is the norm, whereas in a matriarchial society one might expect all members (including males) to consider the perspective of their adversary

Highlight (Yellow) | Location 13046

The Cyc was a little slower to disengage, however. She stayed where she was, maintaining visual lock on Einstein’s epicanthic folds. “What’s your name?” Einstein asked her. “Sonar Taxlaw!” shouted Donno. “Come!” “Now you know it,” she said.

Highlight (Yellow) | Location 13071

“The Pingers!” Einstein called out, as if it were obviously the next topic. “Sonar Taxlaw said they were sea people—before Donno shut her up,” Ty said. “Do you think he beats her?” Einstein asked. It was such an emotional can of worms that Ty considered it carefully before answering. Once in his life, before the war, he had fallen for a girl as quickly as Einstein had for Sonar Taxlaw. That one brief experience with stupid blind love sufficed to make it possible for him to acknowledge its reality and respect its power. “I think,” he said, “that their society is comfortable with corporal punishment to the point where what keeps people like her in line is the fear of it. Not the reality.

Highlight (Yellow) | Location 13077

there’s nothing you can do about it and that if you do so much as look sideways at Donno he will kill you. But you can probably get away with small gestures of kindness toward the Cyc—assuming you’re ever allowed near

her again. If you show her too much favor she will be punished. If you touch her, we're all dead." "Why?" "Because this is one of those cultures that is psychotic about female reproductive organs."

Highlight (Yellow) and Note | Location 13081

Now, let's get back to the Pingers. Name mean anything to you?" "No. You?" said Einstein. Ty's peroration had affected him terribly and reduced him to monosyllables. "I have a vague recollection," Ty said, "but I would have to look it up to be sure."

Sonar taxlaw... (sub)Maritime law, pinger=submarine

Highlight (Yellow) | Location 13095

Ariane and Marge would be reaching a Red habitat around now. For according to the inexorable laws of orbital mechanics, transfer time to geosync was always about twelve hours. He wondered if they were going to the Red capital of Kyoto, or to some military habitat, or

Highlight (Yellow) | Location 13110

there were many others who might have been chosen in his stead—ones better suited to leading an expedition and making first contact with what to the Spacers was an alien race from another planet. No, Ty had been chosen because of where he worked and who owned it. Very old money was behind the Crow's Nest. And enough of it that its Owners didn't mind losing some every month to keep the place going. It was a kind of eleemosynary institution, created to serve not culture and not dukh, but a thing called the Purpose. And if Ty kept working there for another few decades, perhaps one of the Owners would sit him down one day in the Bolt Hole and deign to tell him what exactly the Purpose was.

Highlight (Yellow) | Location 13130

None of the scouts who went out looking for Bard and Beled returned. It was beginning to raise the question of who was really holding whom hostage. The ones who'd gone missing had friends, parents, and children who soon became desperate to know what had become of them and began asking awkward questions of those in command.

Highlight (Yellow) | Location 13170

THAT THESE DROP PODS WERE OF MILITARY DESIGN WAS OBVIOUS from the way they came in: fast. Each had a set of vanes, mounted near the top, that sprang out when it was a couple of thousand meters above the surface, slowing its fall. But not until the pod was just a few tens of meters above the ground did its retro-rockets come on: not just one but a circular array of thumb-sized solids that created a cylindrical piston of fire on which the pod eased to a stop, coming to light on a tripod of buglike legs that deployed themselves at the last possible moment and absorbed the shock of contact.

Highlight (Yellow) and Note | Location 13244

Even at a distance it was obvious that the standard bearer and the one escorting Marge were Aretaics, which was to say, Aïdans of the first line of descent, presumably conceived as competitors to the children of Eve Dinah. They were tall and long-maned, with magnificent noses and excellent posture.

Red

Highlight (Yellow) | Location 13253

Two of the members of the peloton were musicians. At a certain point one of them began to beat a drum that was harnessed to his midsection, and another began to play a melody on a shiny horn. Ty was convinced he'd heard it in the Epic somewhere, but it took Bard to place it. "'Bread of Heaven,'" he said. "It's what Rufus and company were singing when they welded themselves in." "Also known as 'Guide Me, O Thou Great Jehovah,' or in the original Welsh, 'Cwm Rhondda,'" added Sonar Taxlaw. "Fuck, these people are good!" Ty exclaimed.

Highlight (Yellow) | Location 13406

The requirement for a steel-spined authoritarian culture was obvious. Any power structure one of whose main goals was to prevent humans from fucking each other at will had to be extremely formidable. Had these people been living in, for example, the agricultural paradise of the Nile Delta, they might have been able to get away with some mazy religious dogma as the basis for that system. But instead they had been trapped within a large machine that would kill them all if allowed to go on the fritz, and so they had been obliged to develop a culture in which engineering became their dukh. Their finite supply of tungsten, stockpiled by wise Rufus, had to be stretched and husbanded so that their descendants thousands of years in the future would be able to manufacture lightbulbs to grow plants to make food and air. And so on and so forth in every particular of how these people lived their daily lives. Thirty people—the Ten, the Nineteen, and the One—were, at any given time, Cycs. Another thirty were toiling as their apprentices. Others played specific roles such as breeder mom, glassblower, acupuncturist, filament winder, potato nurturer, pump fixer. Structurally, culturally, it was very like a Bronze Age theocracy, but without any trace of God or the supernatural. To that point it was not radically different from the subcultures of many First and Second Millennium space habitats, which—at least for a little while—gave Ty the idea that he could get a quick handle on Digger culture. But that fantasy soon evaporated.

Highlight (Yellow) | Location 13428

All of that had occurred more than a thousand years ago. The Hole had grown to the point where it could support a population of two thousand; then, around 4700 when the atmosphere had become breathable, they'd been able to take it up to ten thousand. All still beneath the surface, however, since there'd been little for them above it. At some point the Committee—which was what they called their ruling council—must have become aware that vast numbers of humans were living in space and actively prosecuting the TerReForm. They could simply have walked out onto the surface and sent out some kind of an SOS at that point. Instead they had made a positive decision to conceal themselves, to hide their spoil dumps, to shun communication with the Spacers. The central question, then, was why they had made such a choice. Sonar Taxlaw wasn't much help in explaining it. When Ty or the others asked questions, she offered nonresponsive answers that told of a subterranean culture in which such things were never spoken of.

Highlight (Yellow) | Location 13437

having made that decision, the Committee would have to explain it, justify it, and perpetuate it by painting the Spacers as alien mutants, and furthermore by cultivating a finely developed sense of racial grievance against the cowards who had run away and abandoned them. All of which had been on vivid display during the brief and disastrous conversation between Doc and the Digger contingent.

Highlight (Yellow) | Location 13568

the Gnomon,” Arjun informed him. Then he spelled the word out. “What does it mean?” “It used to be the thing that stood up in the middle of a sundial, to cast the shadow. Aligned with the Earth’s axis.”

Highlight (Yellow) | Location 13885

Instead of which she found herself gazing at, and being gazed at by, the tall Moiran woman, who had silently drawn almost to within arm’s length of her. Kathree jumped to her feet and nearly fell over. “Kath Amalthova Three,” the woman said, “I am Cantabrigia Barth Five.” Five. Wow. “You must have seen some crazy shit,” Kathree said. “I hope for your sake you have taken a set in your current form.” Cantabrigia Five made a tiny movement of her golden eyes, by way of acknowledging the remark, but let it pass without comment. “I am what amounts to the commanding officer here,” she said. It was neither the craziest nor the least crazy thing Kathree had heard lately, so she took it impassively. By outward appearances, Cantabrigia Five was a video journalist. But it made sense that, in a world where no police or military action could be judged successful unless it looked good to ordinary persons watching it on video screens, she was also a general.

Highlight (Yellow) | Location 14114

Since then, well-meaning members of the group had approached her to offer their sympathies. All they’d wanted to talk about was Doc and Memmie, and what a shock it must have been for Kath to lose them so suddenly. Implicit was that Kath had gone epi because of their deaths. A reasonable-seeming assumption. But wrong. It had happened, rather, in the moment when the old man had attacked her and she had fought back. Doc, at the time, had still been alive, and Memmie, though mortally wounded, had still been breathing. So Kath Two had actually been the first member of the Seven to die. Anyway, she was now the sort of girl who shot people. Useful to know.

Highlight (Yellow) | Location 14209

“Part of you wishes to join the battle. That’s commendable, but I need you—the Purpose needs you—here.” “Fine. You have my attention,” Ty said. Inappropriately, irrelevantly, he was wondering how old this woman was. Epigenetic shifts could roll back many of the visible effects of aging. At least one Moiran, Jamaica Hammerhead Twelve, had lived to the age of two hundred.

Highlight (Yellow) | Location 14371

The Pinger reached out, gently took Ty’s wrist, and plucked at the black stuff of his dry suit. Ty responded by mimicking the same gesture on the slick skin of the Pinger’s arm. Both of them laughed.

Highlight (Yellow) | Location 14392

“Anyway, you were saying about the seventeen photos?” “Yes. Most of them were of a type referred to, in those days, as selfies. Now, technically, this was a criminal violation of military secrecy. Very strange given that Cal was otherwise so attentive to duty.” “Yes,” said Ty, casting his mind back to scenes from the Epic. “I remember Eve Ivy agonizing about that when Eve Julia ordered Cal to nuke Venezuela.”

Highlight (Yellow) | Location 14443

“What’s the Purpose? People keep mentioning it.” “I wish I knew.” “Very funny, but you know what I’m talking about. Roskos Yur mentioned it. Cantabrigia Five mentioned it. Purpose with a capital P.” “My answer remains the same,” Ty said. “No one has ever told me. I have to make guesses, based on what I see from people who act like they know what it is.” “People like the owners of your bar?” “Evidently.” “And what is your guess?” Sensing another pair of eyes on him, Ty glanced over toward Deep, who was chewing vigorously, trying to reduce a stubborn wad of seaweed to submission. But he seemed to be following the conversation. Ty shrugged. “Humans have always—” He was about to say deluded themselves but didn’t want to make a poor impression on Deep. “—preferred to believe that there was a purpose to the universe. Until the moon blew up, they had theories. After Zero, the theories all seemed kind of stupid. Fairy tales for coddled children. No one thought about the big picture for a few thousand years. We were all scrambling to survive. Like ants when their nest has been destroyed. On those rare occasions when we thought about the big picture, it wasn’t really that big—Red versus Blue or what have you. There was surprisingly little thinking about the Agent. Where it came from. Whether it was natural or artificial, or even divine.” Einstein, the Cyc, and Deep were all nodding as if to say Go on, go on! But he had nothing to go on with. “Some people—some Red, some Blue, and some ambiguous folks like the Owners of my bar—maybe even some of those kind of people”—he nodded at Deep—“seem to think they know something.” “Do they?” asked Sonar Taxlaw. “I have no idea,” said Ty. “But from what I’ve seen, they’re not stupid. Even if they are—” He paused, groping for words. “Even if they are,” Einstein repeated, “what?!” “It’s a way—the Purpose is a way—of saying there’s something bigger than this crap we’ve spent the last week of our lives dealing with.” “Red versus Blue crap?” “Yes. And even though no one is sharing anything with me—yet—I like the feeling of that. People who claim they are motivated by the Purpose end up behaving differently—and generally better—than people who serve other masters.” “So it is like believing in God.” “Maybe yes. But without the theology, the scripture, the pigheaded certainty.”

Highlight (Yellow) | Location 14480

You’re right about the Seven. That’s gone. Our Ivyn died. You took his place.” Einstein cackled nervously. “I’m no replacement for Doc!” “You don’t have to replace him. Not in that sense. But look what you did. You made first contact with these guys.” He nodded at Deep. “And first contact of another kind with the Diggers.” Both Einstein and Sonar Taxlaw blushed deeply. “The Cyc came along and replaced Memmie. That’s not a traditional Seven. But if we can pry Beled and Kathree apart, and if we can round up a Julian and a Camite who don’t hate each other, we’ll have ourselves a Nine. The first Nine ever assembled.”

Highlight (Yellow) | Location 14490

“I’ll bet I could talk the Owners out of a few free rooms, for such a momentous occasion.” “Are you really going to ask them?!” Sonar exclaimed. “Nah. As an ancient saying has it, it’s easier to ask forgiveness than

permission. You are all welcome at the Crow's Nest." Ty looked over at Deep. "Just go easy on the cold baths, man. The plumbing in that place has seen better days, and I'm the only one who knows how to fix it."

Highlight (Yellow) | Location 14497

THE PREMISE OF THIS BOOK CAME TO ME CIRCA 2006, WHEN I WAS working part-time at Blue Origin and became interested in the problem of space debris in low Earth orbit.

Highlight (Yellow) | Location 14500

During the same period I had also become aware of the immense amount of usable matter present in near-Earth asteroids. Thus by late 2006 I had come up with the basic premise of Seveneves.

Highlight (Yellow) | Location 14548

In the version that Arthur read, the flivver carrying Kath Two made its final insertion to geosynchronous orbit using a plain old-fashioned rocket burn. Arthur objected to that, not on technical but on aesthetic grounds. This finally pushed me over the brink into using an idea I had been carrying around in the back of my head for a while: having the flivver rendezvous with the end of a cracking whip. The scientific literature on this topic, though sparse, dates back to the Victorian era. The earliest technical reference that I have been able to find on the physics of moving chains is a paper by John Aitken during the 1870s, though he attributes some of its content to his friends the Thomson brothers, William (later Lord Kelvin) and James. Aitken's work lay fallow until the 1920s, when it was picked up and used by M. Z. Carrière in a paper about the physics of whips. Subsequent work published by W. Kucharski (1940) and R. Grammel and K. Zoller (1949) filled out the picture. It is an interesting, underexplored topic in classical physics. I talked about it in a sparsely attended lecture at the Oxford Union in June of 2014, and have intentions of publishing more about it, but nothing definite as of this writing (December 2014).
